

Automotive engine management control IC for small engines

Features

- Supply voltage from 6 V to 18 V
 - Basic functionality guaranteed down to 4.5 V
- 5V regulator up to 300 mA with thermal shutdown protection in current limitation condition
- 5V tracking regulator up to 40 mA and short to battery protection
- 2 channels injectors drivers
 - Parallel and serial driving
 - Output internally clamped to 60 V
 - Minimum overcurrent at 2.7 A
 - Ron 0.6 Ω worst case (at Tj = 150 °C)
- 3 relay drivers
 - 2 with parallel and serial driving, 1 with serial driving
 - Output internally clamped to 45 V
 - Minimum guaranteed output current 1 A
 - Ron 1.5 Ω worst case (at Tj = 150 °C)
- Tachometer driver
 - Parallel and serial driving
 - Minimum guaranteed output current 25 mA
 - Ron 5 Ω worst case (at Tj = 150 °C)
- Low side driver (LED)
 - Serial driving
 - Output internally clamped to 45 V
 - Minimum guaranteed output current 70 mA
 - Ron 20 Ω worst case (at Tj = 150 °C)
- O2 sensor heater
 - Parallel and serial driving
 - Output internally clamped to 45 V
 - Minimum guaranteed output current 3 A
 - Ron 0.55 Ω worst case (at Tj = 150 °C)
- Stepper motor driver
 - Parallel driving
 - Minimum guaranteed output current 500 mA - full step
 - Ron 2.6 Ω worst case on the diagonal (at Tj = 150 °C)
- Protected high side driver
 - 100 mA min. Current limitation threshold
- Full diagnosis by SPI
 - Injector driver: OL, STG, OC
 - Relay and LED drivers: OL, STG, OC
 - O2 sensor heater: OL, STG, OC
 - Tachometer: OL, STG, OC
 - Stepper motor driver: OL, STG, STB, OC
 - general diagnostic: over-temperature
- Protection for STB, STG(for stepper motor drivers and tracking regulator)

- Self-configuring variable reluctance sensor interface(VRS)
- Microcontroller reset logic
- Serial interface
 - SPI 24-bit frame
 - ISO9141 interface (K-Line)

Description

UN67C is a device realized in Automotive BCD technology, able to provide the full set of power supplies and signal preprocessing peripherals needed to control a 2 Cylinder internal combustion Engine for Low End Application (e.g. small motorcycle, K-car, General gasoline engine, etc.).

UN67C integrates a 5V 300 mA main voltage regulator, a 5V 40 mA tracking regulator for sensor supply.

The two channels injector drivers, one O2 sensor heater, two relay drivers can be controlled both with parallel input and with SPI interface. One additional relay driver and the LED driver are controlled by SPI. The stepper motor driver is designed for a double winding coil motor, used for engine idle speed control.

Low side drivers implement SR control to minimize emission.

A protected 50 mA high side driver is provided.

A Variable Reluctance Sensor interface allows the connection to a commercial magnetic pick-up, allowing the indirect measurement of internal combustion engine crank angle. A K-line (standard ISO-9141 compatible) is provided as data communication interface.

The K-line block has slew-rate off function to provide faster baud rate.

All functionalities are fully protected and provide complete diagnostics via a 24-bit SPI interface. An overall protection against over temperature is provided as well.

The device is suitable for automobile emissions standards (China IV and Europe V) system.

The device is available in ETQFP48 package with exposed pad for power dissipation optimization.

Part number	Package	Marking	Packing
UN67CRIID	ETQFP48	UN67C	2500/Tape & Reel

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1 Block diagram and pin description

1.1 Block diagram

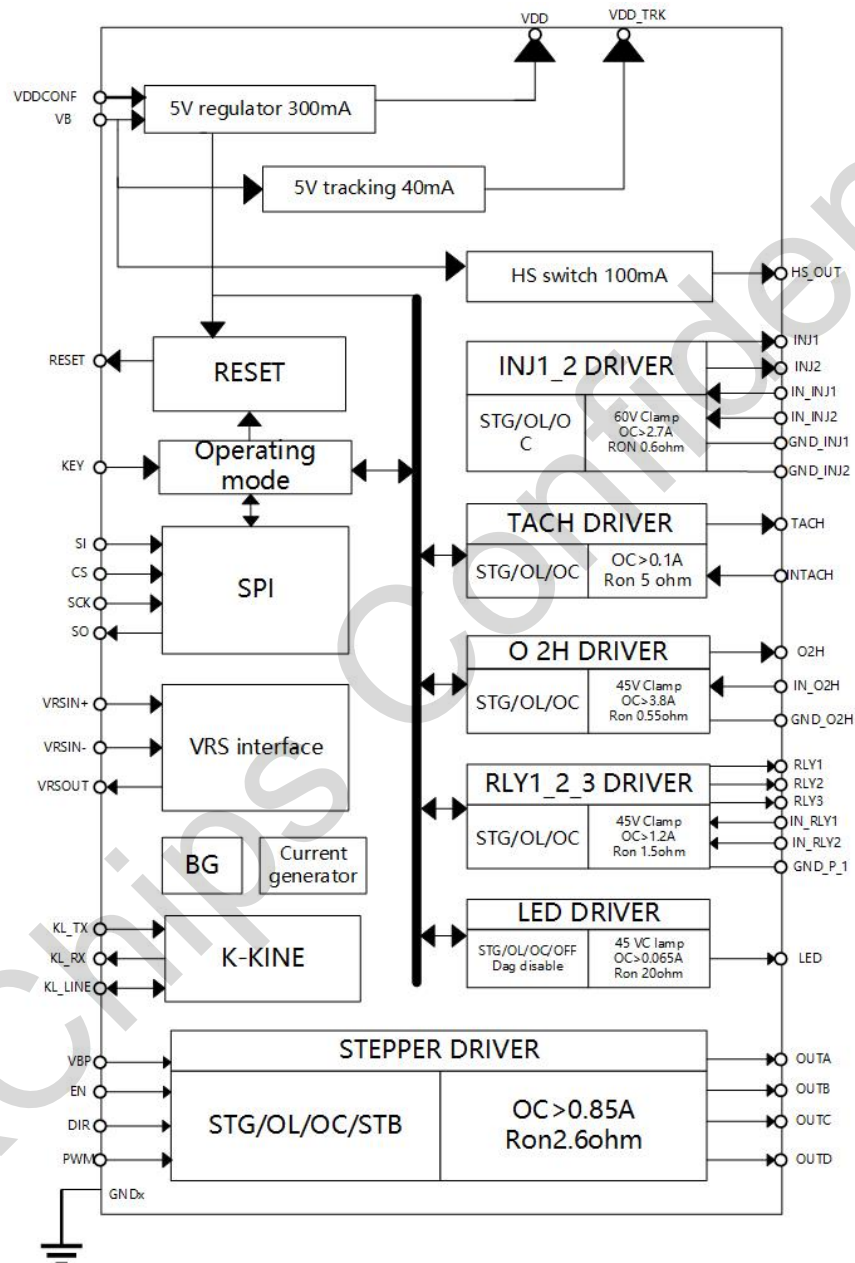


Figure 1-1 Block diagram

1.2 Pin description

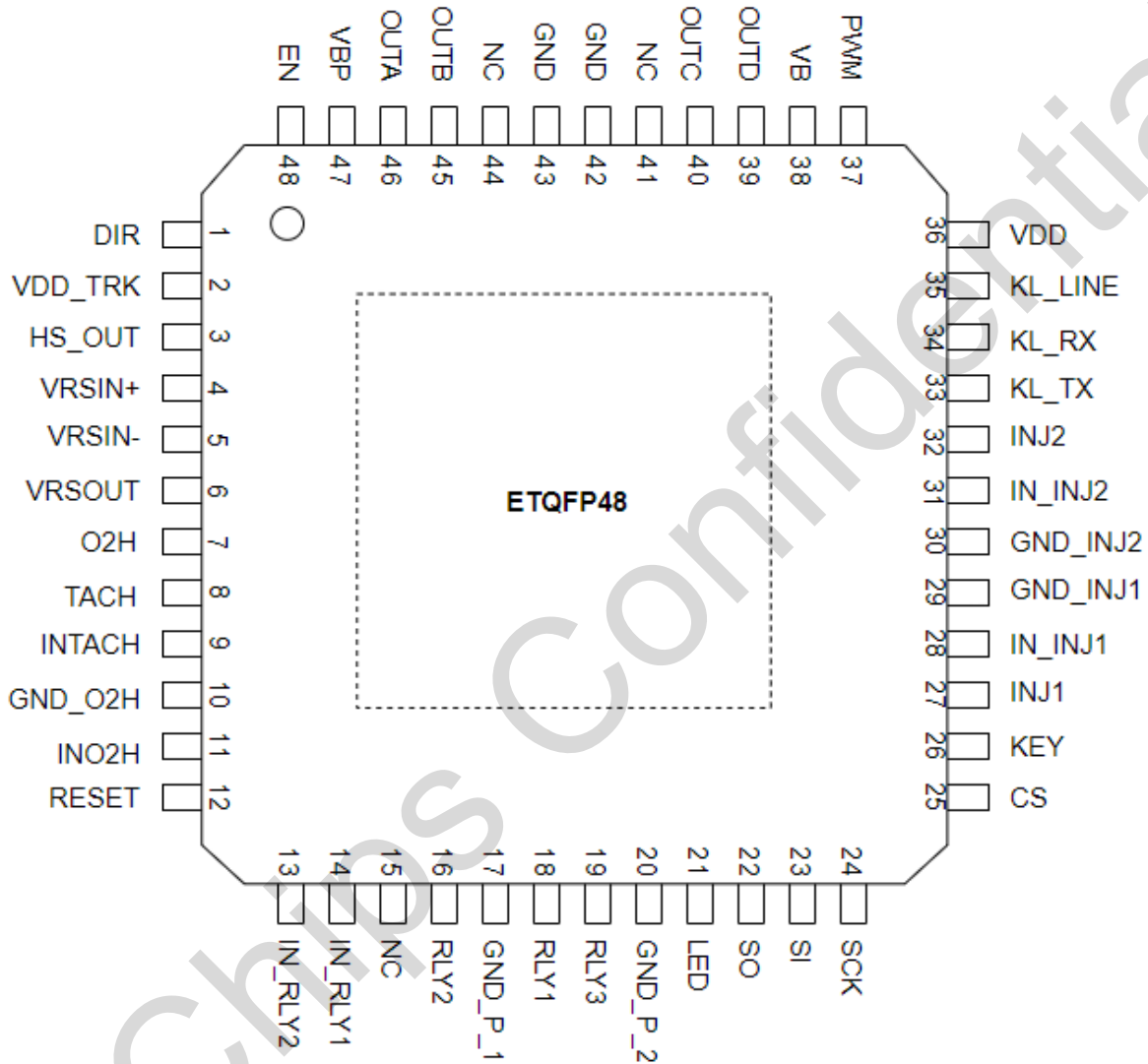


Figure 1-2 PIN OUT(48pin ETQFP48)

Table 1-1 Pin function

Pin#	Pin name	Description	I/O Type	Class
1	DIR	Logic input to set stepper motor direction	I	SIGNAL
2	VDD_TRK	Tracking voltage regulator output	O	PWR
3	HS_OUT	High side switch output	O	PWR
4	VRSIN+	VRS positive differential input	Iss	SIGNAL
5	VRSIN-	VRS negative differential input	I	SIGNAL
6	VRSOUT	VRS output	O	SIGNAL
7	O2H	O2 sensor heater output	O	PWR
8	TACH	Tachometer driver output	O	PWR

9	IN_TACH	Tachometer driver input	I	SIGNAL
10	GND_O2H	O2 sensor heater ground	GND	PWR
11	IN_O2H	O2 sensor heater input	I	SIGNAL
12	RESET	Reset signal to the micro	O	SIGNAL
13	IN_RLY2	Relay 2 parallel control input	I	SIGNAL
14	IN_RLY1	Relay 1 parallel control input	I	SIGNAL
15	NC	Not connected	-	-
16	RLY2	Relay 2 driver output	O	PWR
17	GND_P_1	Power ground relay 1-2	O	PWR
18	RLY1	Relay 1 driver output	O	PWR
19	RLY3	Relay 3 driver output	O	PWR
20	GND_P_2	Power ground for RLY3 and LED	GND	PWR
21	LED	LED driver output	O	PWR
22	SO	SPI data out	O	SIGNAL
23	SI	SPI data in	I	SIGNAL
24	SCK	SPI serial clock	I	SIGNAL
25	CS	SPI chip select	I	SIGNAL
26	KEY	Key signal	I	SIGNAL
27	INJ1	Injector 1 driver power output	O	PWR
28	IN_INJ1	Injector 1 driver input command	I	SIGNAL
29	GND_INJ1	Injector 1 ground	GND	PWR
30	GND_INJ2	Injector 2 ground	GND	PWR
31	IN_INJ2	Injector 2 driver input command	I	SIGNAL
32	INJ2	Injector 2 driver power output	O	PWR
33	KL_TX	K-Line TX digital IN	I	SIGNAL
34	KL_RX	K-Line RX digital OUT	O	SIGNAL
35	KL_LINE	K-Line	I/O	PWR
36	VDD	5V voltage regulator output	O	PWR
37	PWM	Logic input to set Stepper Motor Speed	I	SIGNAL
38	VB	Chip battery supply	S	PWR
39	OUTD	Output bridge 2	O	PWR
40	OUTC	Output bridge 2	O	PWR
41	NC	Not connected	-	-
42	GND	Analog and power ground	GND	PWR
43	GND	Analog and power ground	GND	PWR
44	NC	Not connected	-	-
45	OUTB	Output bridge 1	O	PWR
46	OUTA	Output bridge 1	O	PWR
47	VBP	Chip battery supply for stepper	S	PWR
48	EN	Logic input to enable stepper motor	I	SIGNAL
49	Pad	Exposed PAD	GND	PWR

2 Electrical specifications

2.1 Operating range

The device may not operate properly if maximum operating conditions are exceeded.

Table 2-1 Operating conditions

Parameter	Value	Unit
VB/VBP	supply voltage 6 to 18	V
I/O logic	0 to VDD	V
Stepper motor outputs	-0.3 to VB, VBP	V
Low side outputs	-0.3 to clamp voltage	V

Supply voltage

- Below 3.9 V the device is in a safety state (internal circuitries are on but all the outputs are off).
- From 3.9 V to 5.5 V (Functionalities during Crank phase):
 - Reset function; $VDD > 3.3\text{ V}$ (rds-on state) $IVDD = 100\text{ mA}$; $3.3\text{ V} < VDD_TRK < VDD$ (rds-on state);
 - Low-sides, K-Line, H-Bridge OFF if Reset = 0; SPI not available, internal registers reset if Reset = 0;
 - All Diagnosis disabled if Reset = 0; VRS function limited ($V_{diff\text{ max}} = 1000\text{ mV}$)
- From 5.5 V to 6 V (low battery):
 - All the functions are granted with the following degraded parameters; $VDD > 4.510\text{ V}$; Tracking error $< 100\text{ mV}$ ($I_{load} = 40\text{ mA}$, rds-on state).
- From 6 V to 18 V: normal operating range
- From 18 V to VB_off:
 - All the functions are granted with increased power dissipation and no reset is asserted during transient.
- From VB_off to 40 V (internal circuitries are on but all the outputs are off):
 - The device is on and in a safety state.

2.2 Absolute maximum ratings

Maximum ratings are absolute ratings; exceeding any one of these values may cause permanent damage to the integrated circuit

Table 2-2 Absolute maximum ratings

Parameter	Condition	Min	Max	Unit	Param Num
DC supply	VB, VBP	-0.3	40	V	2.2.1
I/O pin ⁽¹⁾	-	-0.3	40	V	2.2.2
I/O low voltage digital pins ⁽²⁾	-	-0.3	$VDD+0.3$	V	2.2.3
I/O power pins voltage range ⁽³⁾	-	-0.3	Clamp voltage	V	2.2.4
TACH pin	-	-0.3	40	V	2.2.5
OUTA-D	-	-0.3	40	V	2.2.6

KEY pin	To be protected with R _{key} to limit sourced/ sinked current to ± 5 mA in dc conditions and ± 20 mA during transients(ISO-pulses on battery line)	-0.3	40	V	2.2.7
VRSIN- / VRSIN+	Max current [20 mA] to be limited with external resistors	-0.3	VDD+0.3	V	2.2.8
VDD_TRK pin	-	-2	40	V	2.2.9
K_Line pin	-	-16	40	V	2.2.10
Maximum voltage shift between GND pins	PIN GND, GND_O2H, GND_P_1-2, GND_INJ1-2,GNDA,GNDP	-0.3	0.3	V	2.2.11
HS_OUT pin		-0.3	40	V	2.2.12
I/O power pins ⁽³⁾ maximum energy (single pulse, max. current)	Injector drivers		50	mJ	2.2.13
	O2 sensor heater drivers		60	mJ	2.2.14
	Relay		25	mJ	2.2.15
	LED		3.5	mJ	2.2.16
I/O power pins ⁽³⁾ maximum energy (continuous pulse, max. current, 36 million pulses with T = 100ms)	Injector drivers		18	mJ	2.2.17
	O2 sensor heater drivers		22	mJ	2.2.18
	Relay		8	mJ	2.2.19
	LED		0.2	mJ	2.2.20
Reverse current through O2H output without supply voltage ⁽⁴⁾	Static (room temperature, max reverse diode voltage 1.5 V)		2.5	A	2.2.21
Reverse current through INJ1-2 output without supply voltage ⁽⁴⁾	Static (room temperature, max reverse diode voltage 1.5 V)		2.2	A	2.2.22
Reverse current through RLY1-3 output without supply voltage ⁽⁴⁾	Static (room temperature, max reverse diode voltage 1.5 V)		1.5	A	2.2.23
Reverse current through TACH output without supply voltage ⁽⁴⁾	Static (room temperature, max reverse diode voltage 1.5 V)		0.5	A	2.2.24
Reverse current through LED output without supply voltage ⁽⁴⁾	Static (room temperature, max reverse diode voltage 1.0 V)		50	mA	2.2.26

1. Pins are VDD, DIR

2. Pins are CS, SCK, SI, SO, VROUT, RESET, INO2H, PWM, EN, INTACH, INRLY1-2, KL_TX, KL_RX

3. Pins are O2H, INJ1-2, RLY1-2-3, LED

4. Reverse battery connection, parameter not tested for info only

Table 2-3 ESD protection

Item	Condition	Min	Max	Unit	Param Num
All pins ⁽¹⁾⁽²⁾	HBM	-2	2	kV	2.3.1
All pins	CDM (values for corner pins in brackets)	-500/(-750)	500/(750)	V	2.3.2
Pins to connector ⁽³⁾	HBM	-4	4	kV	2.3.3

1. OUTA-D, TACH, O2H, L, INJ1-2, RLY1-2-3 vs. GNDP2, GNDO2: -1.5 / 1.5 kV

2. OUTA-D, TACH, O2H, L, INJ1-2, RLY1-2-3 vs. GNDP1: -1 / 1 kV

3. Pins are OUTA-D, TACH, O2H, INJ1-2, KEY, RLY1-2-3, LED, VDD_TRK all GND connected together. The device is AEC-Q100 compliant

2.3 Latch-up test

According to JEDEC 78 class 2 level A.

2.4 Temperature ranges and thermal data

Table 2-4 Temperature ranges and thermal data

Symbol	Parameter	Min	Max	Unit	Param Num
Tamb	Operating temperature (ECU environment)	-40	85	°C	2.4.1
Tj	Operating junction temperature	-40	150	°C	2.4.2
Tstg	Storage temperature	-40	150	°C	2.4.3
Tot	Thermal shut-down temperature	155	200	°C	2.4.4
Othys	Thermal shut-down temperature hysteresis	10	-	°C	2.4.5
Tflt_ot	Thermal shut-down filter time	20	40	us	2.4.6
Rth j-amb	Thermal resistance junction-to-ambient (1)	-	21	°C/W	2.4.7
Rth j-case	Thermal resistance junction-to-bottom case	-	6	°C/W	2.4.8

1、 with 2s2p PCB thermally enhanced.

3 Electrical characteristics

3.1 Supply electrical characteristics

VB = 6 V to 18 V, Tamb = -40 °C to 85 °C.

Table 3-1 Supply electrical characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
VB	Operating supply voltage range	-	6	-	18	V	3.1.1
VB_off	Vbat switch off threshold voltage	-	29	32	35	V	3.1.2
VB OVh	Overvoltage threshold hysteresis	-	0.5	-	2	V	3.1.3
VB UVL	Undervoltage disable LOW threshold	-	3.9	-	4.5	V	3.1.4
VB UVh	Undervoltage threshold hysteresis	-	0.3	-	1	V	3.1.5
IVB(dis)	Standby current from VB, VBP	VB = VBP = 13 V, device disabled, KEY < 0.7 V	-	-	120	μA	3.1.6
IVB	Quiescent current	VB = VBP = 13 V, outputs floating	-	-	20	mA	3.1.7
fint_clk	Internal clock reference	Application info	-	5.6	-	MHz	3.1.8
Tdgc_VB_OV	VB overvoltage shut-down filter time	Guaranteed by scan	22.5	30	37.5	μs	3.1.9
Tdgc_VB_UV	VB undervoltage shut-down filter time	Guaranteed by scan	0.8	1	1.2	μs	3.1.10

3.2 Key electrical characteristics

Table 3-2 Key electrical characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
V_Key_L	Input level low	-	-	-	1.5	V	3.2.1
V_Key_H	Input level high	-	3.3	-	-	V	3.2.2
V_Key_Hyst	Input voltage hysteresis	-	0.5	-	1.8	V	3.2.3
R_Key	Internal pull down	-	50	150	300	kΩ	3.2.4
T_key_deglitch	Key input filter time	Guaranteed by scan	26	-	40	μs	3.2.5
T_key_delay	Maximum delay time from Key to regulator enable	Time from key rising edge to 20% VDD rising edge	-	-	300	μs	3.2.6

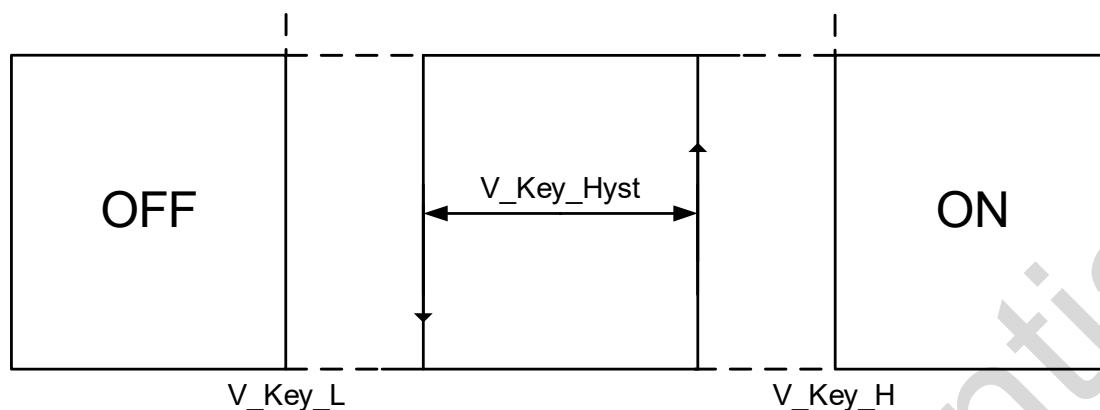


Figure 3-1 Input threshold

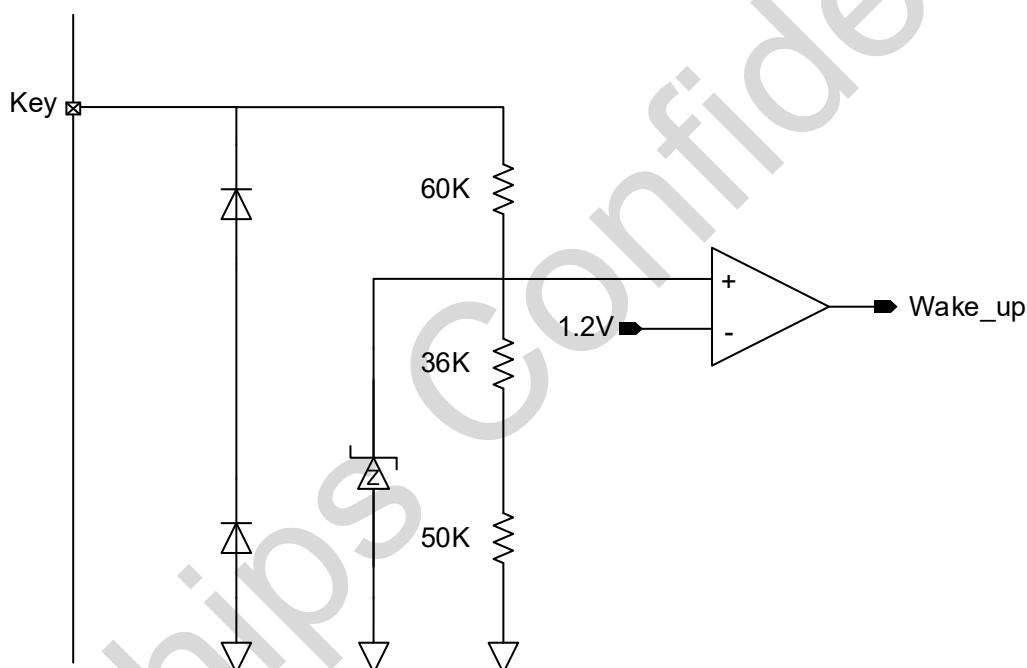


Figure 3-2 Key block diagram

3.3 Digital pins characteristics

Table 3-3 Digital pins characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
Vin_L	Input level low	-	-	-	0.3*VDD	V	3.3.1
Vin_H	Input level high	-	0.7*VDD	-	-	V	3.3.2
Vh_in	Input voltage hysteresis	-	0.1	-	-	V	3.3.3
R_pull	Internal pull-down/pull-up ⁽¹⁾⁽²⁾	-	50	150	250	kΩ	3.3.4
I_pull_down	Active pull-down	-	10	-	100	μA	3.3.5

1. Pins with pull-down: EN, DIR, SI, SCK, PWM, INTACH, ININJ1-2, INO2H, IN_RLY1-2;

2. Pins with pull-up: CS, KL_TX;

Table 3-4 Digital output pins characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
Vout_L	Output level low ⁽¹⁾⁽²⁾	I sink= 2mA	-	-	0.4	V	3.4.1
Vout_H	Output level high ⁽¹⁾	I source= 2mA	VDD-0.5	-	-	V	3.4.2

1. Pins with push-pull stage and tristate condition: DO;

2. Pins with open drain output: RESET, VRSOUT.

3.4 VDD voltage regulator

Table 3-5 VDD output electrical characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
VDD	Output voltage	-	4.9	5	5.1	V	3.5.1
Ln_vdd	Line regulation	VB = 6 V to 18 V Iload = 150 mA	-25	-	25	mV	3.5.2
Ld_vdd	Load regulation	VB = 13 V Iload = 5 mA to 300 mA	-25	-	25	mV	3.5.3
Vdd_OS	Max overshoot	Recovery from ISO pulse stimuli on battery line (guaranteed by design)	-	-	5.5	V	3.5.4
Vdd_SR	Voltage slew-rate at power-on	Cload = 4.7 μ F	2	-	25	V/ms	3.5.5
Idd	Load current	-	5	-	300	mA	3.5.6
Idd_max	Current limitation	Output short to 4 VDD-1V	350	-	600	mA	3.5.7
Idd_STG	Short to ground current limitation	Output shorted to GND	350	-	700	mA	3.5.8
PSRR	Power supply rejection ratio	Sin wave @ 1 kHz 1 Vpp; VB = 13 V; Iload = 5 mA to 300 mA	40	-	-	dB	3.5.9
Vdr5	VB – Vdd, Vdd dropout voltage	VB = 5 V; Iload = 300 mA	0.1	-	0.75	V	3.5.10
Tdcg_VDD	VDD thermal shutdown filter time	Guaranteed by scan	22.5	30	37.5	μ s	3.5.11

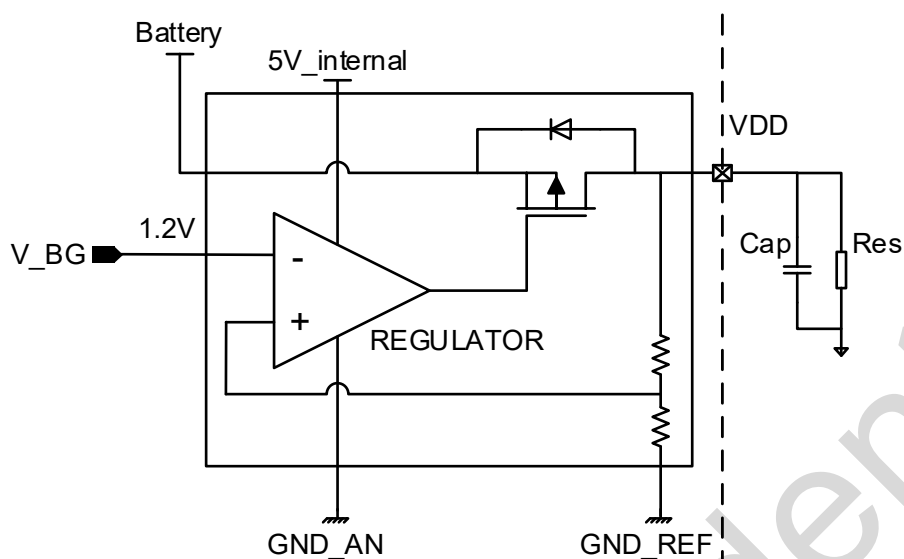


Figure 3-3 VDD regulator block diagram

3.5 Reset function electrical characteristics

Table 3-6 Reset function electrical characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
VUV_LO	Output low voltage	$1.8 < VDD < V_{th_UV}$, $I_{reset} = 2 \text{ mA}$	-	-	0.6	V	3.6.1
IUV_LO	Reset current capability	$1.8 < VDD < V_{th_UV}$, $V_{reset} = 0.6 \text{ V}$	2	-	-	mA	3.6.2
I _{lk}	Leakage current	$VUV_reset = 4.5 \text{ V}$	-	-	1	μA	3.6.3
V _{th_UV}	VDD under voltage low threshold		4.5	-	$VDD - 150 \text{ mV}$	V	3.6.4
V _{th_UVTh}	VDD under voltage high threshold		4.5	-	$VDD - 50 \text{ mV}$	V	3.6.5
V _{th_UV} HYS	VDD under voltage hysteresis		50	-	-	mV	3.6.6
Td_UV_rst	Power on UV reset delay	Guaranteed by scan	17	22	30	ms	3.6.7
TfUV_reset	UV reset filter	$VDD < V_{th_UV}$, Guaranteed by scan	25	50	75	μs	3.6.8

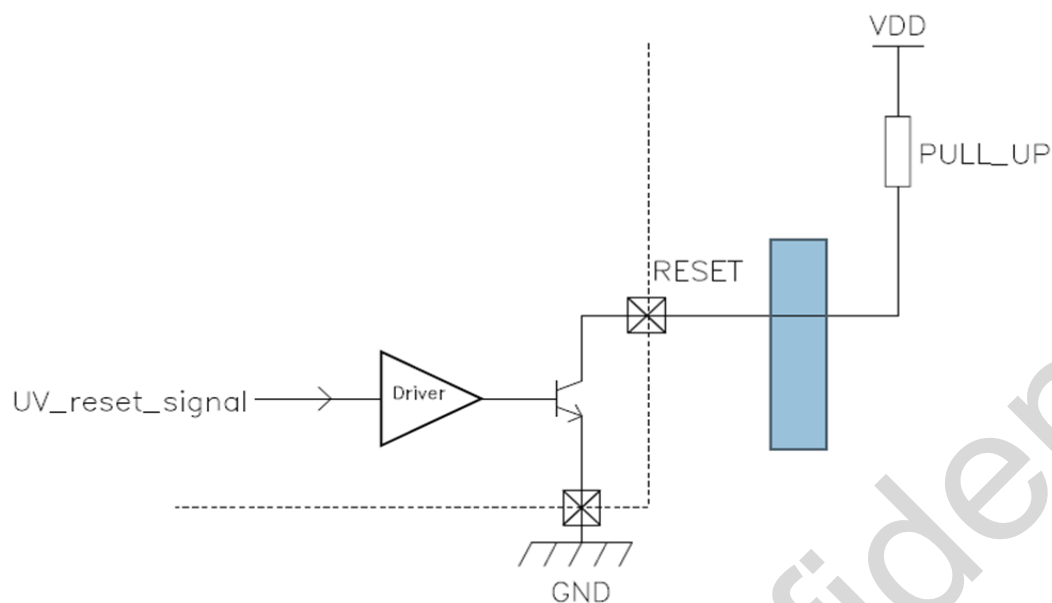


Figure 3-4 Reset

3.6 List of figures VDD Tracking voltage regulator

Table 3-7 VDD_TRK output electrical characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
DVddtrk	Output voltage tracking error	VB = 6 V, Itrk= 1 to 40 mA	-15	-	15	mV	3.7.1
Vshort	Tracking output short circuit voltage range	-	-2	-	VB	V	3.7.2
Itrk_max	Output current limitation	Output short to VDD-1V	50	-	100	mA	3.7.3
Itrk_sb	Tracking output short to battery current (limited by the regulator)	Output shorted to VB 18V	-	-	10	mA	3.7.4
Idd	Load current	-	1	-	40	mA	3.7.5
Ln_vdd_trk	Line regulation	VB= 6 V to 18 V - Iload= 40 mA	-15	-	15	mV	3.7.6
Ld_vdd_trk	Load regulation	VB= 13 V Iload = 1 to 40 mA	-15	-	15	mV	3.7.7
PSRR	Power supply rejection ratio	Sin wave @ 1 kHz 1 Vpp VB = 13 V, Iload = 1 to 40 mA	40	-	-	dB	3.7.8

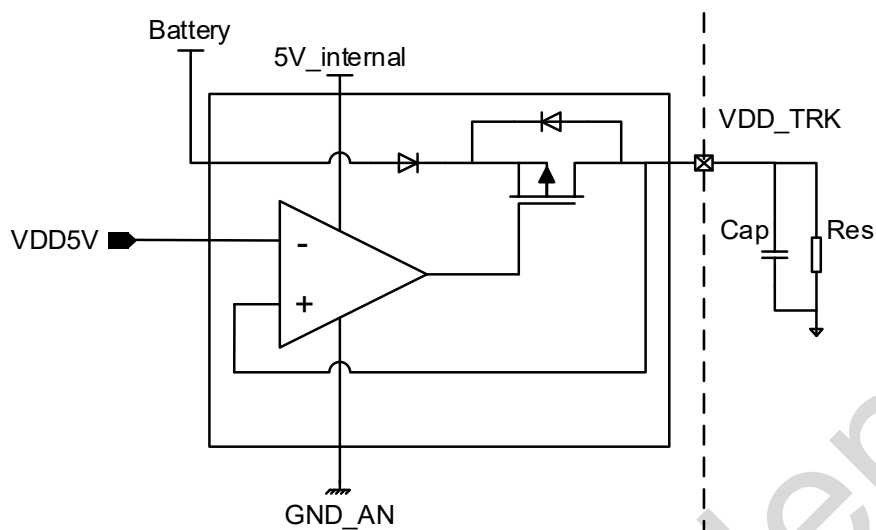


Figure 3-5 tracking regulator block diagram

3.7 High side switch

Table 3-8 HS_OUT output electrical characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
Ron_hs	Ron	I _{hs} = 50 mA	-	-	14	Ω	3.8.1
I _{hs_max}	Current limitation	V _B = 13.5 V	100	-	400	mA	3.8.2

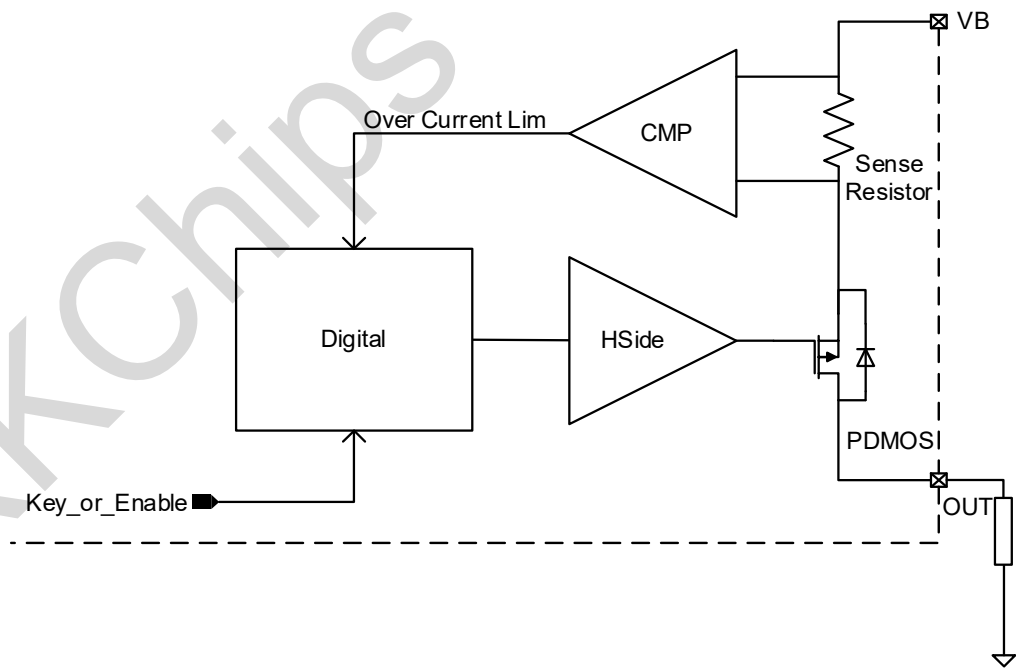


Figure 3-6 High-side driver block diagram

The High side switch is intended as a protected battery and is directly controlled by key input.

3.8 Injector driver

Table 3-9 Injector driver electrical characteristic

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
I _{max}	Output current	-	-	-	2.2	A	3.9.1
I _{oc}	Overcurrent threshold	-	2.7	-	5	A	3.9.2
V _{DS}	Output clamping voltage	I = 2.2 A.	50	-	65	V	3.9.3
R _{on}	On resistance	I = 2.2 A	-	-	0.6	Ω	3.9.4
I _{lk_off}	Leakage current	V _{out} = 18 V, Key = 0 V	-	-	10	μA	3.9.5
I _{lk_on}	Pull-Down diagnosis current	V _{out} = 18 V, Key = 5 V	-	-	100	μA	3.9.6
T _{on_OUTx}	Propagation Delay from CMD (serial or parallel) edge to 80% output OUTx voltage	V _B =14V, R _{load} =15 Ω, C _{load} =10nF	-	-	6	us	3.9.7
T _{off_OUTx}	Propagation Delay from CMD (serial or parallel) edge to 20% output OUTx voltage	V _B =14V, R _{load} =15 Ω, C _{load} =10nF	-	-	6	us	3.9.8
V _{OL}	Open load output voltage	Driver in OFF condition	2.3	2.5	2.7	V	3.9.9
V _{diagth_H} ⁽¹⁾	Diagnostic high threshold	Driver in OFF condition	2.7	3	3.3	V	3.9.10
V _{diagth_L} ⁽¹⁾	Diagnostic low threshold	Driver in OFF condition	1.8	2	2.2	V	3.9.11
T _{mask}	OFF diagnostic masking time	Guaranteed by scan	0.75	1	1.25	ms	3.9.12
T _{dcg_noise}	OFF diagnostic Deglitch filter time	Guaranteed by scan	35	50	65	μs	3.9.13
T _{dcg}	ON diagnostic Deglitch filter time	Guaranteed by scan	2.8	4	5.2	μs	3.9.14

1. V_{diagth_L} < V_{out} < V_{diagth_H} → Open Load; V_{out} < V_{diagth_L} → Short to GND

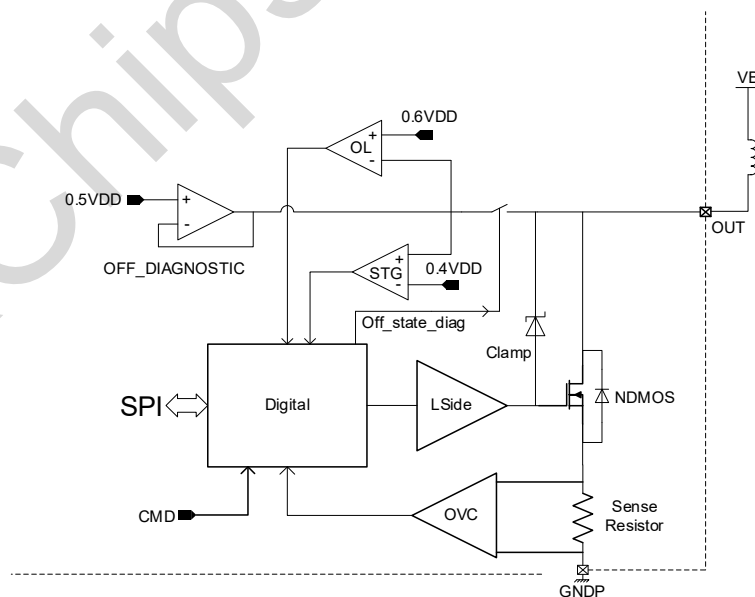


Figure 3-7 Low-side driver block diagram

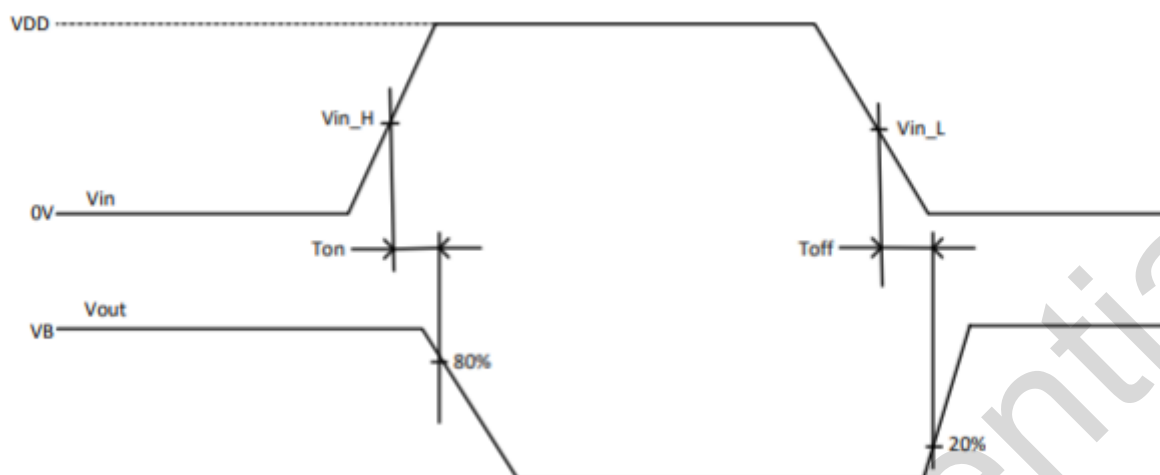


Figure 3-8 Low-side timing diagram (injectors, relays, LED, tach, O2H)

3.9 Relay drivers

Table 3-10 Relay driver characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
I_{max}	Output current	-	-	-	1	A	3.10.1
I_{oc}	Overcurrent threshold	-	1.2	-	2.5	A	3.10.2
V_{DS}	Output clamping voltage	$I = 1\text{ A}$	40	-	50	V	3.10.4
R_{on}	On resistance	$I = 1\text{ A}$	-	-	1.5	Ω	3.10.5
I_{lk_off}	Leakage current	$V_{out} = 18\text{ V}$, Key = 0 V	-	-	10	μA	3.10.6
I_{lk_on}	Pull-Down diagnosis current	$V_{out} = 18\text{ V}$, Key = 5 V	-	-	100	μA	3.10.7
T_{on_OUTx}	Propagation Delay from CMD (serial or parallel) edge to 80% output OUTx voltage	$V_B = 14\text{ V}$, $R_{load} = 68\Omega$, $C_{load} = 10\text{ nF}$			6	μs	3.10.8
T_{off_OUTx}	Propagation Delay from CMD (serial or parallel) edge to 20% output OUTx voltage	$V_B = 14\text{ V}$, $R_{load} = 68\Omega$, $C_{load} = 10\text{ nF}$			6	μs	3.10.9
V_{OL}	Open load output voltage	Driver in OFF condition	2.3	2.5	2.7	V	3.10.10
$V_{diagth_H}^{(1)}$	Diagnostic high threshold	Driver in OFF condition	2.7	3	3.3	V	3.10.11
$V_{diagth_L}^{(1)}$	Diagnostic low threshold	Driver in OFF condition	1.8	2	2.2	V	3.10.12
T_{mask}	OFF diagnostic masking time	Guaranteed by scan	2.63	3.5	4.38	ms	3.10.13
T_{dcg_noise}	OFF diagnostic Deglitch filter time	Guaranteed by scan	35	50	65	μs	3.10.14
T_{dcg}	ON diagnostic Deglitch filter time	Guaranteed by scan	2.8	4	5.2	μs	3.10.15

1. $V_{diagth_L} < V_{out} < V_{diagth_H} \rightarrow$ Open Load; $V_{out} < V_{diagth_L} \rightarrow$ Short to GND

3.10 Tachometer driver

Table 3-11 Tachometer driver electrical characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
I _{oc}	Overcurrent threshold	-	100	-	500	mA	3.11.1
R _{on}	On resistance	I = 25 mA	-	-	5	Ω	3.11.2
I _{lk_off}	Leakage current	V _{out} = 18 V, Key = 0 V	-	-	10	μA	3.11.4
I _{lk_on}	Pull-Down diagnosis current	V _{out} = 18 V, Key = 5 V	-	-	100	μA	3.11.5
T _{on_OUTx}	Propagation Delay from CMD (serial or parallel) edge to 80% output OUTx voltage	V _B = 14V, R _{load} = 270Ω	-	-	6	us	3.11.6
T _{off_OUTx}	Propagation Delay from CMD (serial or parallel) edge to 20% output OUTx voltage	V _B = 14V, R _{load} = 270Ω	-	-	6	us	3.11.7
V _{OL}	Open load output voltage	driver in OFF condition	2.3	2.5	2.7	V	3.11.8
V _{diagth_H} ⁽¹⁾	Diagnostic high threshold	Driver in OFF condition	2.7	3	3.3	V	3.11.9
V _{diagth_L} ⁽¹⁾	Diagnostic low threshold	Driver in OFF condition	1.8	2	2.2	V	3.11.10
T _{mask}	OFF diagnostic masking time	Guaranteed by scan	0.75	1	1.25	ms	3.11.11
T _{dcg_noise}	OFF diagnostic Deglitch filter time	Guaranteed by scan	35	50	65	μs	3.11.12
T _{dcg}	ON diagnostic Deglitch filter time	Guaranteed by scan	2.8	4	5.2	μs	3.11.1

Note: V_{diagth_L} < V_{out} < V_{diagth_H} → Open Load; V_{out} < V_{diagth_L} → Short to GND

3.11 Stepper motor driver

Table 3-12 Stepper motor driver electrical characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
I _{oc}	Overcurrent threshold	ON condition	0.85	-	2	A	3.12.1
R _{dsON}	On resistance HS+LS	I _{out} = 0.5 A, T _j = 150 °C, V _B = 14 V	-	-	2.6	Ω	3.12.2
f _{stepper}	Working frequency	Application info	-	-	20	kHz	3.12.3
V _{out_off(1)(2)(3)}	OUTA_B_C_D output voltage	OUTA short to OUTB; OUTC short to OUTD; Stepper driver disable	2.3	2.5	2.7	V	3.12.4
V _{diagth_H}	Diagnostic high threshold	Driver in OFF condition	2.7	3	3.3	V	3.12.5
V _{diagth_L}	Diagnostic low threshold	Driver in OFF condition	1.8	2	2.2	V	3.12.6
I _{DSS_OUT}	Output leakage current	Driver in OFF condition	-	-	10	μA	3.12.7
t _{scvb}	Over current switch_off time	Guaranteed by scan	-	-	25	μs	3.12.8
t _{rb}	Rise output time	V _B = 12 V, R _I = 39 Ω	-	-	15	μs	3.12.9
t _{fb}	Fall output time	V _B = 12 V, R _I = 39 Ω	-	-	15	μs	3.12.10
t _{rb-a}	Rise output time	T _{amb} = 25 °C, V _B = 12 V, R _I = 39 Ω	-	-	10	μs	3.12.11
t _{fb-a}	Fall output time	T _{amb} = 25 °C, V _B = 12 V, R _I = 39 Ω	-	-	10	μs	3.12.12

t_{pHLb}	Turn-off in/out delay time	$V_B = 12\text{ V}, R_I = 39\ \Omega$	-	-	15	μs	3.12.13
t_{pLHb}	Turn-off in/out delay time		-	-	15	μs	3.12.14
$V_{reverse_HS}$	Reverse HS diode drop	Driver in OFF condition $I_{injected} = 0.5\text{ A}$	-	-	1.5	V	3.12.15
$V_{reverse_LS}$	Reverse LS diode drop	Driver in OFF condition $I_{sourced} = 0.5\text{ A}$	-	-	-1.5	V	3.12.16
$T_{dgc_step_OFF}$	OFF Diagnostic Deglitch filter time	Guaranteed by scan	93.8	125	156.3	μs	3.12.17
T_{mask}	Masking time	Guaranteed by scan	0.75	1	1.25	ms	3.12.18
$T_{dgc_step_ON}$	ON Diagnostic Deglitch filter time	Guaranteed by scan	15	20	25	μs	3.12.19

1. $V_{outA} < V_{diagth_H}$ and $V_{outB} > V_{diagth_L}$ or $V_{outC} < V_{diagth_H}$ and $V_{outD} > V_{diagth_L} \rightarrow$ No Fault

2. $V_{outA_B_C_D} < V_{diagth_L} \rightarrow$ Short to GND

3. $V_{outA} > V_{diagth_H}$ and $V_{outB} < V_{diagth_L}$ or $V_{outC} > V_{diagth_H}$ and $V_{outD} < V_{diagth_L} \rightarrow$ Open load 4. $V_{outA_B_C_D} > V_{diagth_H} \rightarrow$ Short to Battery

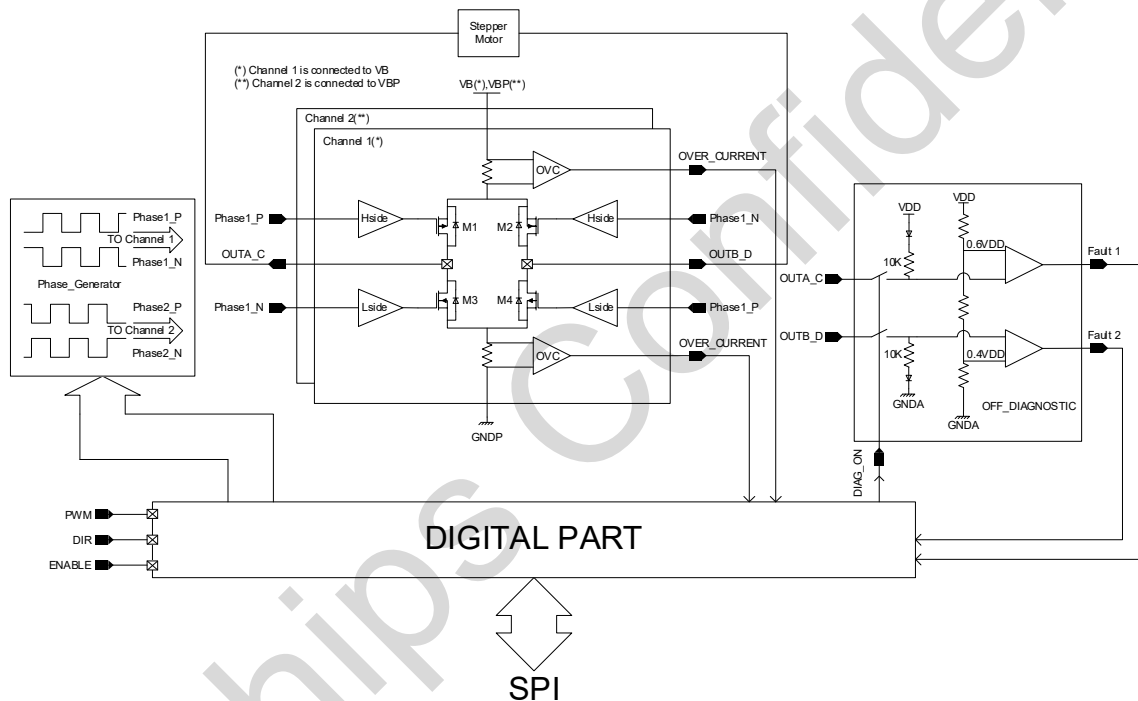


Figure 3-9 Stepper motor driver block diagram

3.12 O2 sensor heater driver

Table 3-13 O2 sensor heater driver characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
R_{dsON}	On resistance	$I_{out} = 3\text{ A}$	-	-	0.55	Ω	3.13.1
V_{DS}	Output clamping voltage	$I_{out} = 3\text{ A}$	40	-	50	V	3.13.2
I_{lk_off}	Leakage current	$V_{out} = 18\text{ V}, \text{Key} = 0\text{ V}$	-	-	10	μA	3.13.3
I_{lk_on}	Pull-Down diagnosis current	$V_{out} = 18\text{ V}, \text{Key} = 5\text{ V}$	-	-	100	μA	3.13.4
T_{on_OUTx}	Propagation Delay from CMD (serial or parallel) edge	$V_B_IN = 14\text{ V}, R_{load} = 8\ \Omega, C_{load} = 10\text{ nF}$			10	μs	3.13.5

	to 80% output OUTx voltage						
Toff_OUTx	Propagation Delay from CMD (serial or parallel) edge to 20% output OUTx voltage	VB_IN=14V, Rload=8 Ω, Cload=10nF			10	us	3.13.6
VOL	Open load output voltage	Driver in OFF condition	2.3	2.5	2.7	V	3.13.7
IOC_L	low Overcurrent threshold		3.8	-	5.2	A	3.13.8
Vdiagth_H ⁽²⁾	Diagnostic high threshold	Driver in OFF condition	2.7	3	3.3	V	3.13.9
Vdiagth_L ⁽²⁾	Diagnostic low threshold	Driver in OFF condition	1.8	2	2.2	V	3.13.10
Tmask	OFF diagnostic masking time	Guaranteed by scan	0.75	1	1.25	ms	3.13.11
Tdcg_noise	OFF diagnostic Deglitch filter time	Guaranteed by scan	28	40	52	μs	3.13.12
Tdcg	ON diagnostic Deglitch filter time	Guaranteed by scan	2.8	4	5.2	μs	3.13.13

1. $V_{diagth_L} < V_{out} < V_{diagth_H} \rightarrow$ Open Load; $V_{out} < V_{diagth_L} \rightarrow$ Short to GND

3.13 LED driver

Table 3-14 LED driver characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
RdsON	On resistance	Iout = 50mA	-	-	20	Ω	3.14.1
VDS	Output clamping voltage	Iout = 10mA	40	-	50	V	3.14.2
Ilk_off	Leakage current	Vout= 18 V, Key = 0 V	-	-	10	μA	3.14.3
Ilk_on	Pull-Down diagnosis current	Vout= 18 V, Key = 5 V	-	-	100	μA	3.14.4
SR_ON	Voltage slew ON State	From 80% to 30% of VOUT, VB =14V, Rload=270 Ω	8		24	V/us	3.14.5
SR_OFF	Voltage slew OFF State	From 30% to 80% of VOUT, VB=14V, Rload=270 Ω	8		24	V/us	3.14.6
Ton_OUTx	Propagation Delay from CMD (serial or parallel) edge to 80% output OUTx voltage	VB =14V, Rload=270 Ω	-	-	6	us	3.14.7
Toff_OUTx	Propagation Delay from CMD (serial or parallel) edge to 20% output OUTx voltage	VB =14V, Rload=270 Ω	-	-	6	us	3.14.8
VOL	Open load output voltage	Driver in OFF condition	2.3	2.5	2.7	V	3.14.9
IOC	Overcurrent threshold		65	-	120	mA	3.14.10
Vdiagth_H ⁽¹⁾	Diagnostic high threshold	Driver in OFF condition	2.7	3	3.3	V	3.14.11
Vdiagth_L ⁽¹⁾	Diagnostic low threshold	Driver in OFF condition	1.8	2	2.2	V	3.14.12
Tmask	OFF diagnostic masking time	Guaranteed by scan	0.75	1	1.25	ms	3.14.13
Tdcg_noise	OFF diagnostic Deglitch filter time	Guaranteed by scan	35	50	65	μs	3.14.14
Tdcg	ON diagnostic Deglitch filter time	Guaranteed by scan	14	20	26	μs	3.14.15

1 $V_{diagth_L} < V_{out} < V_{diagth_H} \rightarrow$ Open Load; $V_{out} < V_{diagth_L} \rightarrow$ Short to GND

3.14 Variable reluctance sensor interface

Table 3-15 Variable reluctance sensor interface electrical characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
VHTH	Input high to low differential threshold voltage at VRSIN+ and VRSIN- nodes	-	-50	-	50	mV	3.15.1
VCM	Common mode operating range at VRSIN+ and VRSIN-	-	0	1.5	3	V	3.15.2
V _{cm_internal}	Internal common mode voltage	Switch off hysteresis current	0.27*VDD	0.3*VDD	0.33*VDD	V	3.15.3
V _{diagth_H} ⁽¹⁾	Diagnostic high threshold	Diagnostic voltage referred to VRSIN-	0.67*VDD	0.75*VDD	0.82*VDD	V	3.15.4
V _{diagth_L} ⁽¹⁾	Diagnostic low threshold		0.18*VDD	0.2*VDD	0.22*VDD	V	3.15.5
I _{diag}	Diagnostic current	INP = open, INN = 10k to gnd; Current from VRSIN- when diagnostic on	35	50	65	μA	3.15.6
R _{internal_common}	Internal common mode resistor	Switch off hysteresis current	50	200	350	kΩ	3.15.7
I _{IB}	Input bias current	VRSIN+ = VRSIN- = 1.5 V	-	-	2	μA	3.15.8
I _{leak}	Output leakage	VROUT = 5 V	-	-	1	μA	3.15.9
V _{CLPH}	Input high clamping voltage	VRSIN+ = VRSIN- = 20 mA	-	5	-	V	3.15.10
V _{CLPL}	Input low clamping voltage	VRSIN+ = VRSIN- = [20 mA]	-1.5	-	-0.3	V	3.15.11
t _{of}	Output fall time	CLOAD= 20 pF, RLOAD = 5 kΩ	-	-	300	ns	3.15.12
t _{of_1nf}	Output fall time	CLOAD= 1nF	-	-	1.5	μs	3.15.13
V _{OUTL}	Output buffer low voltage	I _{sink} = 2 mA	-	-	0.6	V	3.15.14
I _{OUTL}	Output current capability	V _{out} = 0.6 V	2	-	-	mA	3.15.15
t _{prop}	Propagation delay	VRS INM = 0.5 V, INP applied to 1 V to make VRS OUT commuted	0.1	0.45	0.8	μs	3.15.16
T _{dgc_VRS}	OFF Diagnostic Deglitch filter time	Guaranteed by scan	21	30	39	μs	3.15.17

¹ if (VRSIN- > V_{diagth_H}) or (VRSIN- < V_{diagth_L}) then Fault is detected.

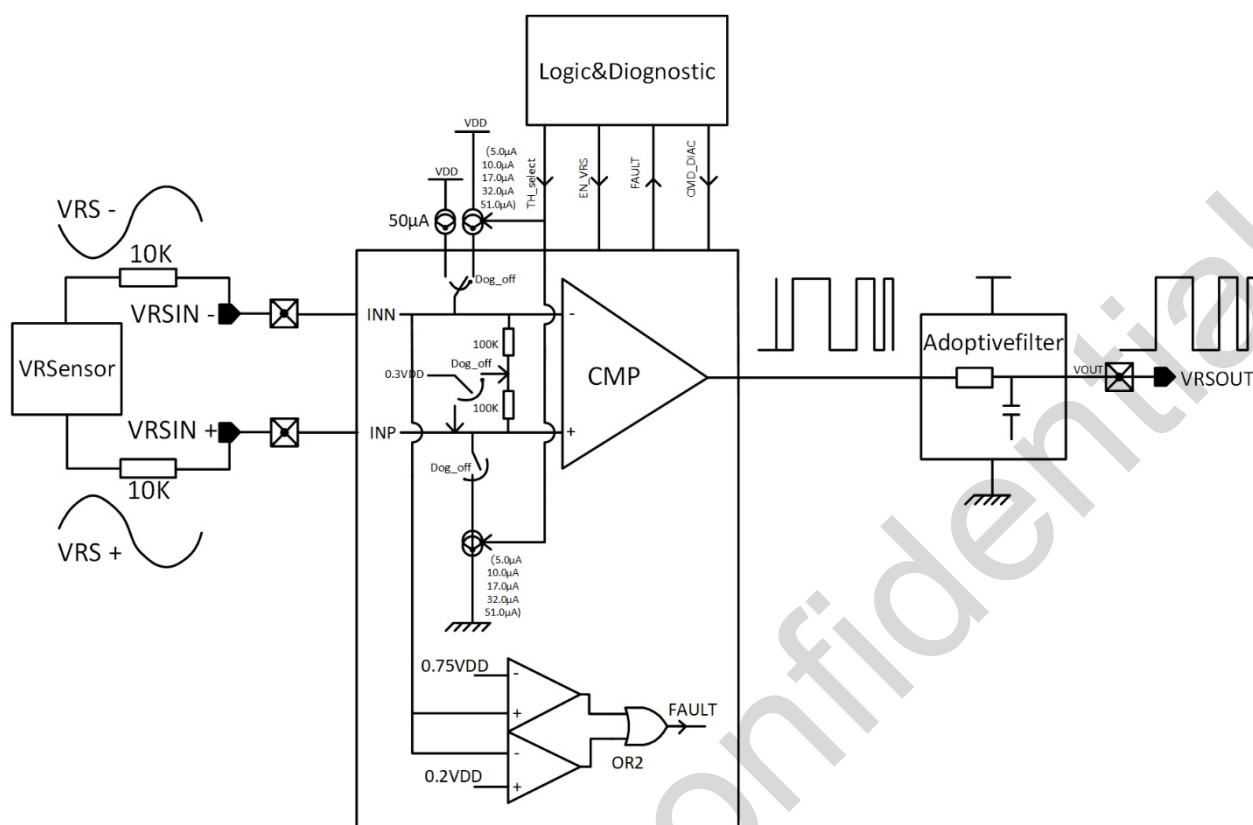


Figure 3-10 VRS block diagram

Table 3-16 Pick Voltage Detector Precision

Symbol	Parameter	Test Condition	Min	Typ	Max	Unit	Param Num
PV1	Pick Voltage	-	600	930	1300	mV	3.16.1
PV2	Pick Voltage	-	1200	1600	1950	mV	3.16.2
PV3	Pick Voltage	-	2000	2300	2650	mV	3.16.3
PV4	Pick Voltage	-	2600	3000	3300	mV	3.16.4

Table 3-17 Hysteresis threshold precision

Symbol	Parameter	Correspondent value on 20KΩ ext. resistor	Min	Typ	Max	Unit	Param Num
HI1	Hysteresis Current	100 mV	3(60mV)	5(100mV)	7(140mV)	µA	3.17.1
HI2	Hysteresis Current	200 mV	7(140mV)	10(200mV)	13(260mV)	µA	3.17.2
HI3	Hysteresis Current	340 mV	13(260mV)	17(340mV)	21(420mV)	µA	3.17.3
HI4	Hysteresis Current	640 mV	26(520mV)	32(640mV)	38(760mV)	µA	3.17.4
HI5	Hysteresis Current	1020 mV	38(760mV)	51(1020mV)	60(1200mV)	µA	3.17.5

1 For a single IC, there is no overlap of parameter PVX (PV1<PV2<PV3<PV4) and HIX(HI1<HI2<HI3<HI4<HI5), which are guaranteed by design

3.15 K-line interface electrical characteristics

Table 3-18 K-Line interface electrical characteristics

Pin	Symbol	Parameter	Test conditions	Min.	Typ.	Max.	Unit	Param Num
KL_TX	ITXsource	Transmitter input source current	KL_TX = GND	10	-	100	μA	3.19.1
	ITXsink	Transmitter input sink current	KL_TX = VDD	-	-	2.1	μA	3.19.2
	IKL_TX	KL_TX Internal Pull-up	KL_TX = GND	-	150	-	kΩ	3.19.3
K_LINE	VKoutL	Transmitter output low voltage	IsinkK_LINE = 35 mA, KL_TX = Low	-1	-	1.5	V	3.19.4
	VKinH	Receiver input high voltage	-	0.7xVB	-	VB	V	3.19.5
	VKinL	Receiver input low voltage	-	-1	-	0.35xVB	V	3.19.6
	VKH	Receiver input hysteresis	-	0.05xVB	-	0.3xVB	V	3.19.7
	IKleak	Receiver leakage current	KL_LINE = VB, KL_TX = High	-	-	1	μA	3.19.8
K_LINE	IKshort	Transmitter short circuit current	KL_LINE = VB, KL_TX = Low	60	-	-	mA	3.19.9
	IKrev	Reverse battery or GND loss current	ENABLE = KEY = VB = 0 V, KL_LINE = -13.5 V	-	-	10	mA	3.19.10
	IKpull-up	KLINE internal pull-up	KL_TX = High	60	-	140	kΩ	3.19.11
	IKuv	Under voltage current	KEY = High, KL_TX = Low, VB = 13.5 V, KL_LINE = -1 V	-	-	1	mA	3.19.12
KL_RX	VRXoutL	KL_RX output low voltage	Isink = 0.4 mA	-	-	0.4	V	3.19.13
KL_TX to K_LINE	Tp_HLT	Transmitter turn-on delay time	CKline = 10nF, RKline = 510Ω	-	-	5	μs	3.19.14
K_LINE	T_fT	Transmitter fall time	CKline = 10nF, RKline = 510Ω	-	-	10	μs	3.19.15
KL_LINE to KL_RX	T_fR	Receiver turn-on delay time	Cload = 20 pF, RPKL_Rx = 2 kΩ	-	-	4	μs	3.19.16
KL_RX	T_fR	Receiver fall time	Cload = 20 pF, RPKL_Rx = 2 kΩ	-	-	2	μs	3.19.17
	T_rR	Receiver rise time	Cload = 20 pF, RPKL_Rx = 2 kΩ	-	-	2	μs	3.19.18
K_LINE	fMax	Max transmission Operating frequency	Application info	-	-	60	kHz	3.19.19

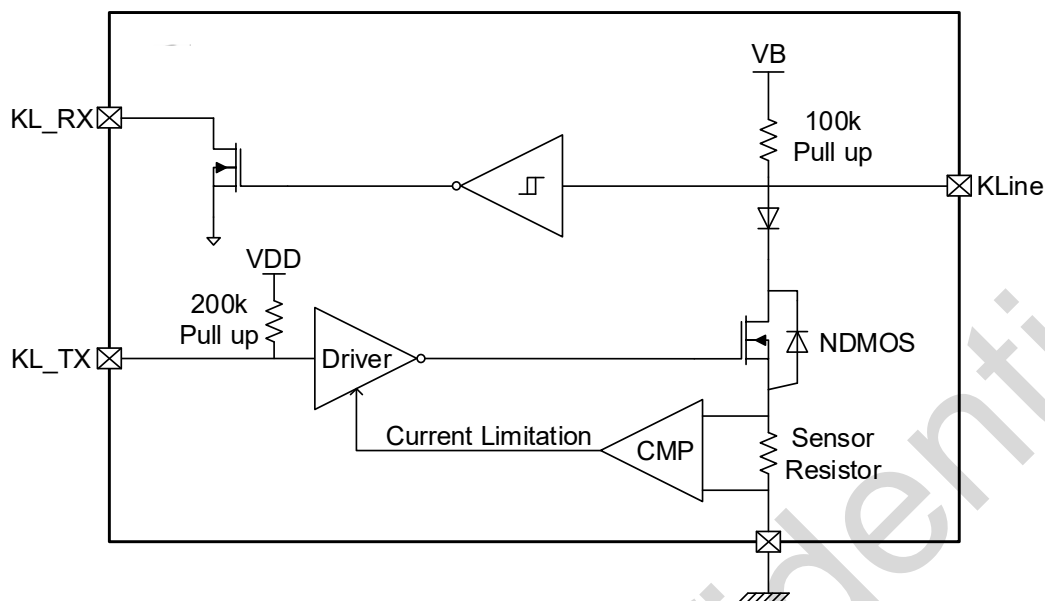


Figure 3-11 K-line block diagram

3.16 SPI interface

Table 3-19 SPI characteristics and timings

Symbol	Parameter	Test conditions	Min	Typ	Max	Unit	Param Num
SICin	Input capacitance	-	-	-	20	pF	3.20.1
SCKCin		-	-	-	20	pF	3.20.2
tSCKCS	Clock inactive time before frame	-	100	-	-	ns	3.20.3
tCSSO	Access time	See ⁽¹⁾	-	-	500	ns	3.20.4
tSODis	Output data (SO) disable time	No Capacitor on SO, See ⁽²⁾	-	-	500	ns	3.20.5
tlead	Channels select (CS) lead time	See ⁽¹⁾	500	-	-	ns	3.20.6
tSCKFSO	Output valid time	See ⁽¹⁾ , @ fCLK= 5.4 MHz	60	-	-	ns	3.20.7
tSOCS	Output data (SO) disable time	No capacitor on SO, see ⁽¹⁾	-	-	500	ns	3.20.8
tSIsetup	Input data (SI) set-up time	@ fCLK= 5.4 MHz	20	-	-	ns	3.20.9
tSIhold	Input data (SI) hold time	@ fCLK= 5.4 MHz	20	-	-	ns	3.20.10
tSCK	CLK period	-	185	-	-	ns	3.20.11
tCSSCK	Clock inactive time after frame	-	600	-	-	ns	3.20.12
tCSN	CS de asserted time	-	600	-	-	ns	3.20.13

1. See Figure 3-12

2. See Figure 3-11

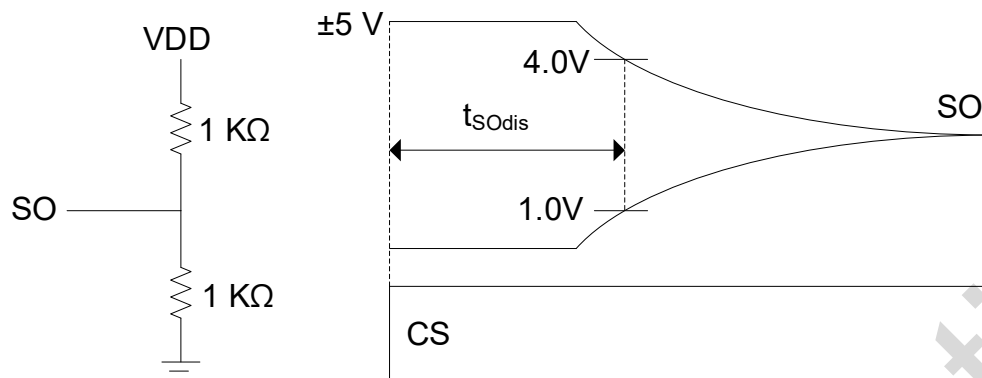


Figure 3-12 SO loading for disable time measurement

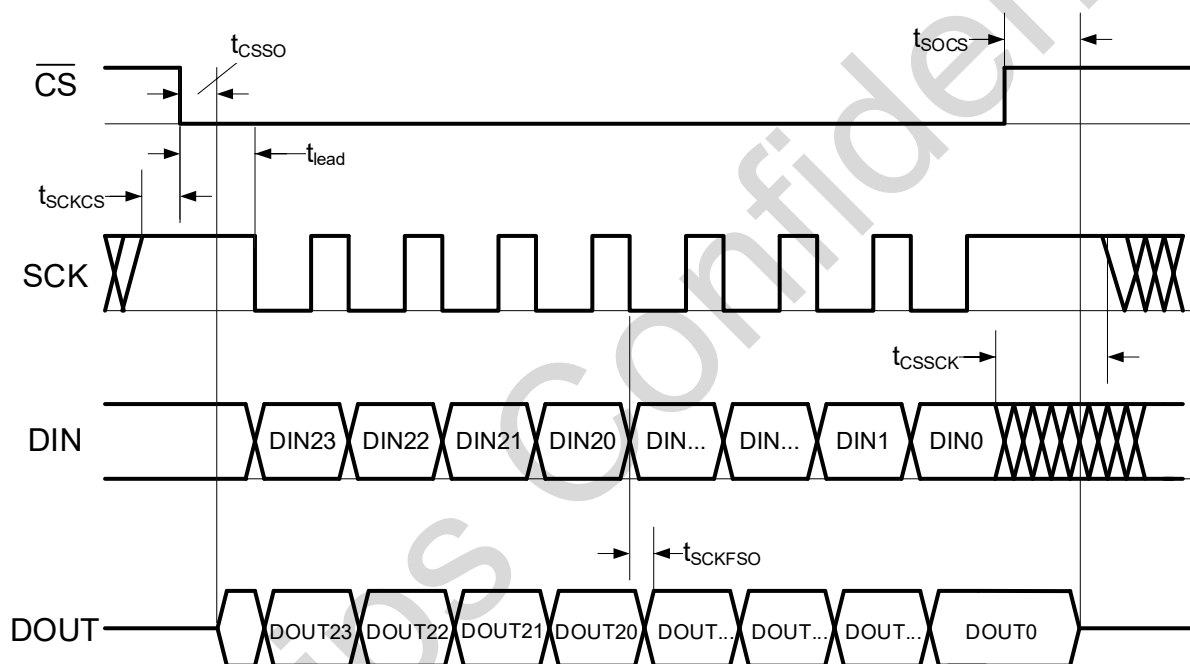


Figure 3-13 SPI timing diagram

4 Functional description

4.1 Chip working conditions

Table 4-1 Outputs working conditions

-	Run mode	VB_OV	VB_UV	Reset	Over current	Over-temperature
VDD regulator	ON	OFF	OFF	ON	Current limitation	OFF if linked with VDD current limitation
VDD_TRK regulator	ON	OFF	OFF	ON	Current limitation	OFF if linked with VDD_TRK current limitation
All LS drivers	ON	OFF	OFF	OFF	Over current switch off	OFF
Diagnostics of all LS drivers	ON	OFF	OFF	ON(1)	-	ON
HS Driver	ON	OFF	OFF	OFF	Current limitation	ON
Stepper Motor Driver	ON	OFF	OFF	OFF	Over current switch off	OFF
K-line Transceiver	ON	OFF	OFF	OFF	Current limitation	ON
VRS	ON	OFF	OFF	OFF	-	ON
SPI	Default	Default	Default	Default	ON	ON

Note: The diagnostic currents and comparator are switched on in reset condition.

4.2 Power up/down sequences

The figures below show the power-on, power-off and time diagram behavior of the device.

When the KEY_IN signal rises up and remains stable for at least $T_{key_deglitch}$, the device goes in ON state, meaning that all voltage regulators and functions are active.

Wake-up is an intermediate status between standby and on mode, with current consumption higher than the standby one.

When Key_IN goes low, device goes in OFF mode.

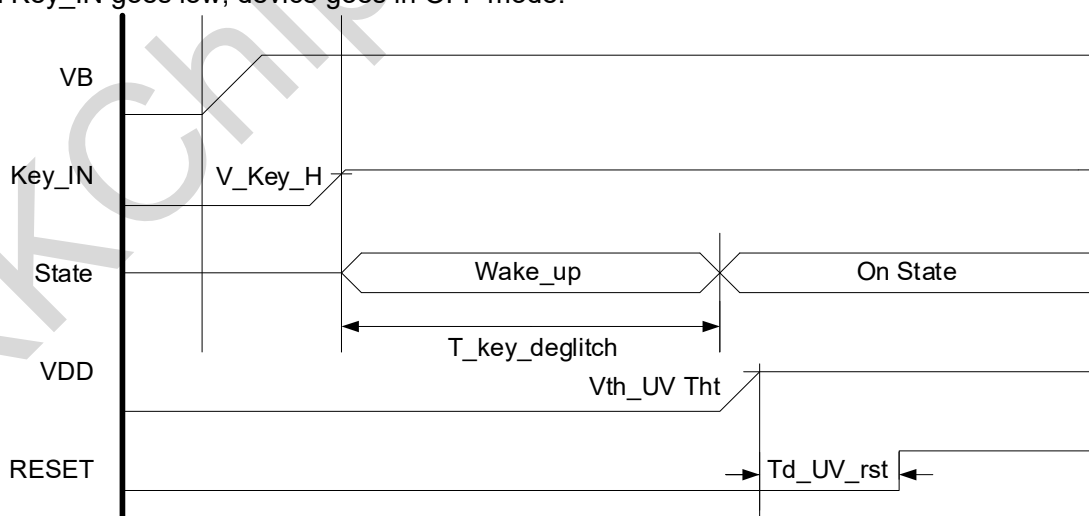


Figure 4-1 Power-up sequence

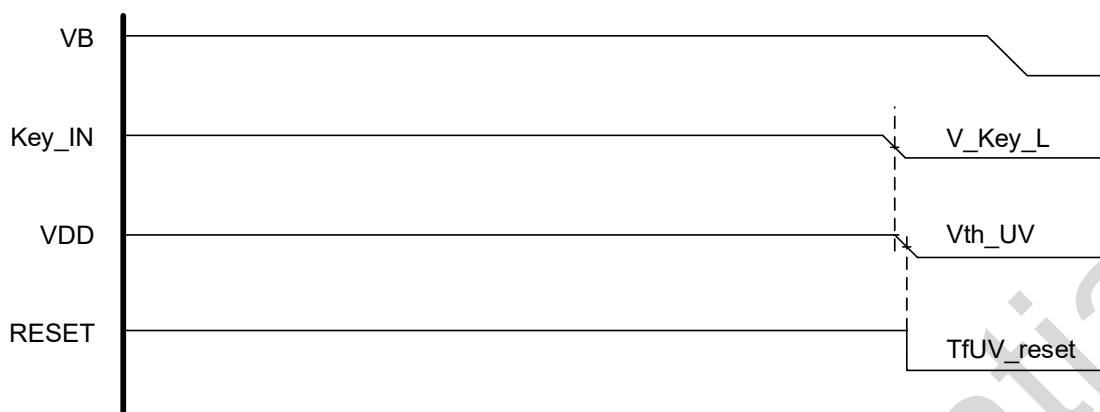


Figure 4-2 Power-down sequence

Reset signal detects a VDD undervoltage longer than TfUV_reset by going to low level. When VDD recovers to normal level Reset signal returns to high level after Power_On_UV_Reset_Delay time (Td_UV_RST). The Reset signal resets all the internal SPI registers to default value.

4.3 SPI

SPI is a standard four wires interface that communicates with a data word of 24 bits. By means of SPI all the channels can be driven in serial way and diagnosis is sent out. Timing of SPI's operations are reported in Figure 3-12. The input data (DIN) is read on the rising edge of the SPI's clock (SCLK), in the same way the output data (DOUT) must be read by the Microcontroller on the SCLK's rising edge.

4.3.1 Data in (DIN)

DIN command is used to turn On/Off internal channels which do not have Parallel Input command, and to clear diagnostic latches.

DIN is decoded at the end of the frame if the integrity checks are passed.

Table 4-2 Data in (DIN) words content

Bit23	Bit22	Bit21	Bit20	Bit19	Bit18	Bit17	Bit16
Mask	LED_0	LED_1	VRS Diag	VRS Hys0	VRS Hys1	VRS Hys2	0
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
INJ1_0	INJ1_1	INJ2_0	INJ2_1	O2H_0	O2H_1	RLY1_0	RLY1_1
Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
RLY2_0	RLY2_1	RLY3_0	RLY3_1	TACH_0	TACH_1	Clear diag	Parity

Data in structure (MSB first)

- **Mask** bit is used to mask serial command for diagnosis only readings on DOUT.
0 read diag. All DIN bits are ignored
1 write, All DIN are transferred into the internal registers.
- Command bits are used to control the output drivers (INJ1-2,O2H,RLY1-2-3,LED, Tach) as described in the following table:

Table 4-3 Data in command bits structure

XXX_0	XXX_1	Description
0	1	Turn-off driver / parallel polarity 0
1	1	Turn-on driver / parallel polarity 1
X	0	No change(the driver will maintain the previous condition)

- **VRS Diag** bit is used active high to enable diagnostic phase of VRS block, the diagnosis can be done only when the phonic wheel is stopped.
- Programmable VRS Hysteresis: VRS hysteresis is programmable in 5 steps according to Table 4-4.
- **Clear Diag**, when set to 1 generates a request to clear those diagnostic flags which are latched.
- In addition **odd parity bit** (that is the last bit of the frame and includes in its calculation the "Don't care" bits) is used for DIN word check together with falling clock edges count.

Table 4-4 Data in VRS hysteresis

Hys0	Hys1	Hys2	Description
0	0	0	auto adaptive hysteresis
1	0	0	Hys VRS=100mV
0	1	0	Hys VRS=200mV
1	1	0	Hys VRS=340mV
0	0	1	Hys VRS=640mV
1	0	1	Hys VRS=1020mV
1	1	1	Not Valid(Hys doesn't change)

4.3.2 Data out (DOUT)

Status flags are sampled and sent out through DOUT pin at each R/W SPI operation. The structure of the 24 bit words is described in Table 4-6. A three bits diagnosis is provided for stepper motor driver, a two bits diagnosis for other drivers. VRS diagnosis is coded as '0' means No Fault, while '1' means Fault. Over temperature warning is coded as '0' means No Fault, while '1' means Fault. The SPI default value is: all bits set to zero.

Table 4-5 Data out (DOUT) words content

Bit23	Bit22	Bit21	Bit20	Bit19	Bit18	Bit17	Bit16
INJ1 Diag0	INJ1 Diag1	INJ2 Diag0	INJ2 Diag1	O2H Diag0	O2H Diag1	RLY1 Diag0	RLY1 Diag1
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
RLY2 Diag0	RLY2 Diag1	RLY3 Diag0	RLY3 Diag1	LED Diag0	LED Diag1	TACH Diag0	TACH Diag1
Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
VRS Diag	Over-temperature	Brdg1 Diag0	Brdg1 Diag1	Brdg1 Diag2	Brdg2 Diag0	Brdg2 Diag1	Brdg2 Diag2

Data out structure

Table 4-6 Two bits diagnosis (normal drivers)

Bit1	Bit0	Fault
0	0	no Fault
0	1	STG(OFF)

1	0	OPL(OFF)
1	1	Overcurrent(On)

Table 4-7 Three bits diagnosis (bridge stage)

Bit0	Bit1	Bit2	Fault
0	0	0	no Fault
1	0	0	STG(OFF)
1	0	1	STB(OFF)
0	1	0	Open Load (OFF)
1	1	0	Over current (On)

4.4 Diagnosis

The device provides a full set of diagnosis; deglitch timings listed below are digital, generated from internal clock and their accuracy is guaranteed by scan patterns and clock measurement.

4.4.1 Thermal shut down

There are 5 temperature sensors:

- OT1 for VDD and VTRK
- OT2 for O2H1 and TACH
- OT3 for RELAY1-3,LED
- OT4 for INJ1-2
- OT5 for Stepper motor driver

When OT1 is higher than Tot for tOT time VDD, VTRK are switched off if they are in current limitation.
When OT1 is lower than junction - Othys for tOT time, the device should return to normal operation automatically.

When OT2 is higher than Tot for tOT time O2H1 and TACH are switched off.
When OT2 is lower than Tot - Othys for tOT time, the channels are able to be switched on again.

When OT3 is higher than Tot for tOT time RELAY1-3and LED are switched off.
When OT3 is lower than Tot - Othys for tOT time, the channels are able to be switched on again.

When OT4 is higher than Tot for tOT time INJ1-2 are switched off.
When OT4 is lower than Tot - Othys for tOT time, the channels are able to be switched on again.

When OT5 is higher than Tot for tOT time Stepper motor driver is switched off.
When OT5 is lower than Tot - Othys for tOT time, the channels are able to be switched on again

Thermal warning information from OT1, OT2, OT3, OT4 and OT5 was communicated by SPI.
Over temperature flag without any latch is present via SPI.

4.4.2 Voltage regulators thermal warning and shutdown

The VDD linear voltage regulator/tracking regulator is shut down when the thermal shutdown

temperature is reached and also the regulator is in current limitation. The shutdown is filtered with Tdgc filter of $30 \mu\text{s} \pm 25\%$. As soon as the over temperature disappears the regulator is switched on again. Over temperature flag without any latch is present via SPI.

4.4.3 Overvoltage shut down

If the VB_off voltage is reached after Tdgc filtering time of $30 \mu\text{s} \pm 25\%$ the device enters a safety state where the main outputs are switched-off. Voltage regulators, all low side channels, stepper motor driver, HS_OUT and KLINE are switched off and reset is asserted. As soon as the battery goes below VB_off minus VB_off_h the device recovers standard operation.

4.4.4 Undervoltage shut down

If the VB_UV voltage is reached after analog Tdgc $1 \mu\text{s} \pm 20\%$ filtering time the device enters a safety state where main outputs are switched-off. Voltage regulators, all low side channels, stepper motor driver, HS_OUT and KLINE are switched off. As soon as the battery rises above VB_UV plus the hysteresis the device recovers normal operation.

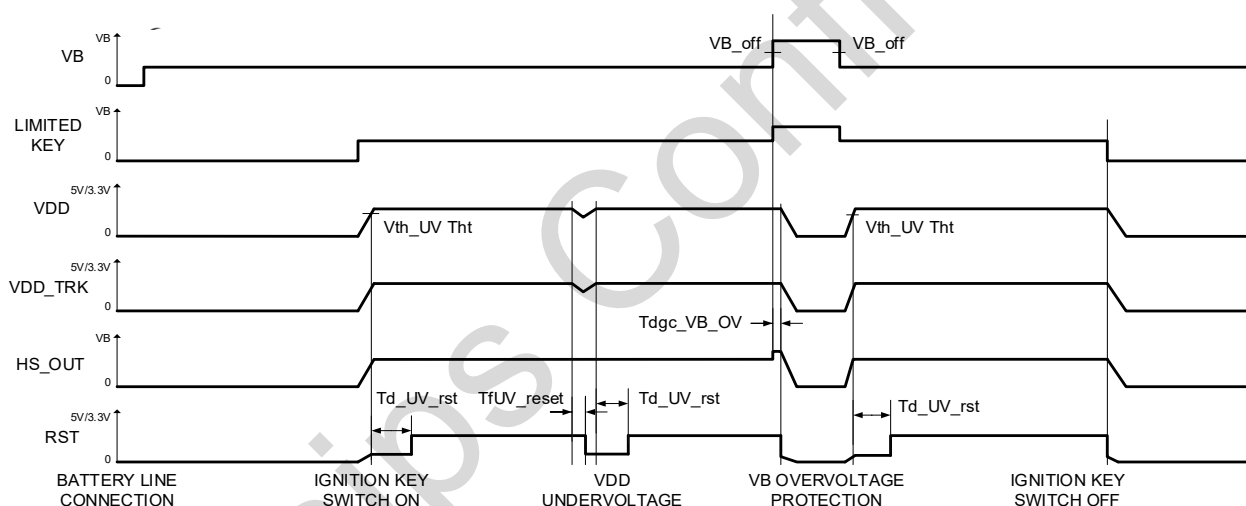


Figure 4-3 An example of under and over voltage time diagram

4.4.5 Low side on/off diagnosis (INJ, RLY's, TACH, O2H, LED)

About low side channels OFF diagnosis, the device issues a masking filter Tmask after channel turning off (falling edge of driving command) to avoid false fault detecting due to output transition from low to high. Tmask is of $1 \text{ ms} \pm 25\%$ for all channels except for the relays, for which Tmask is $3.5 \text{ ms} \pm 25\%$. Once masking time has expired a deglitch filter Tdgc_noise for noise immunity is activated. A fault longer than deglitch time is latched. OFF state diagnostic fault can be overwritten by ON state fault. OFF state fault does not prevent the driver from switching on. The latched fault is cleared on request.

During on-phase if an over current fault occurs the drivers enter in current limitation condition for a digital filtering time Tdgc, then it is switched OFF and the fault is latched. The channel is turned ON

again by input command transition. The latched fault is cleared on request via SPI.

Over current fault has higher priority respect to OFF condition faults.

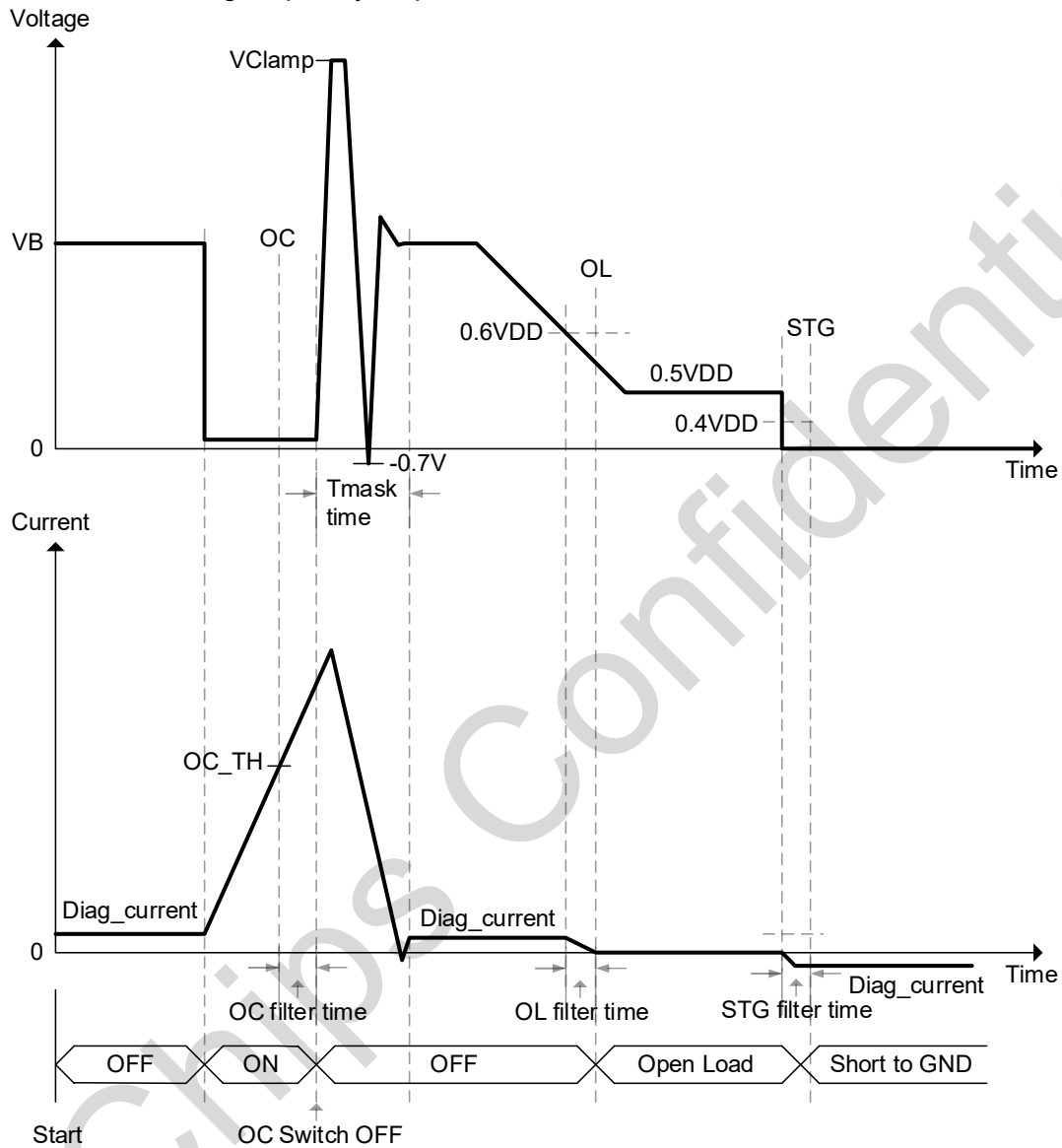


Figure 4-4 Low side driver diagnosis time diagram

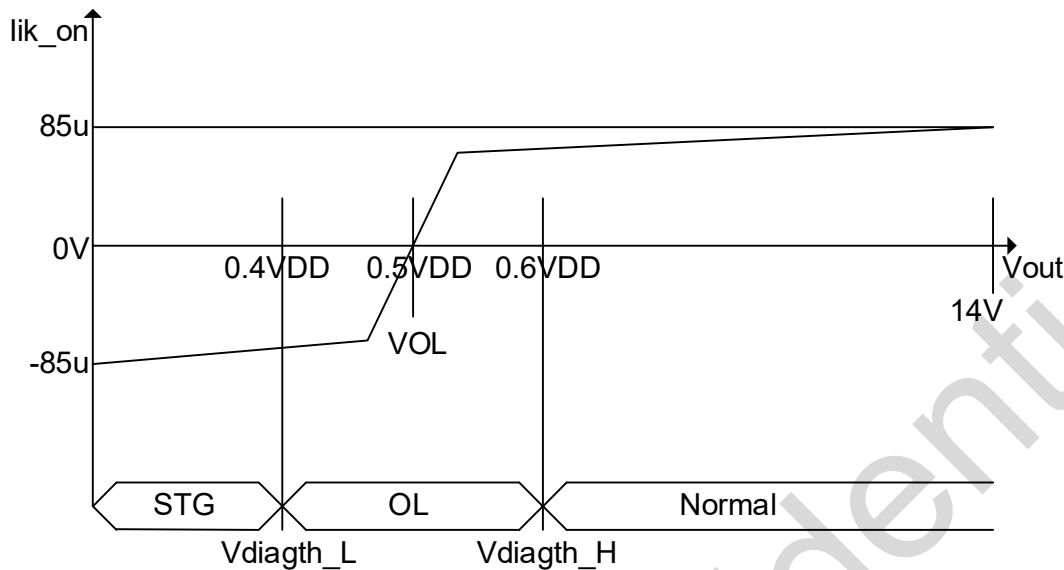


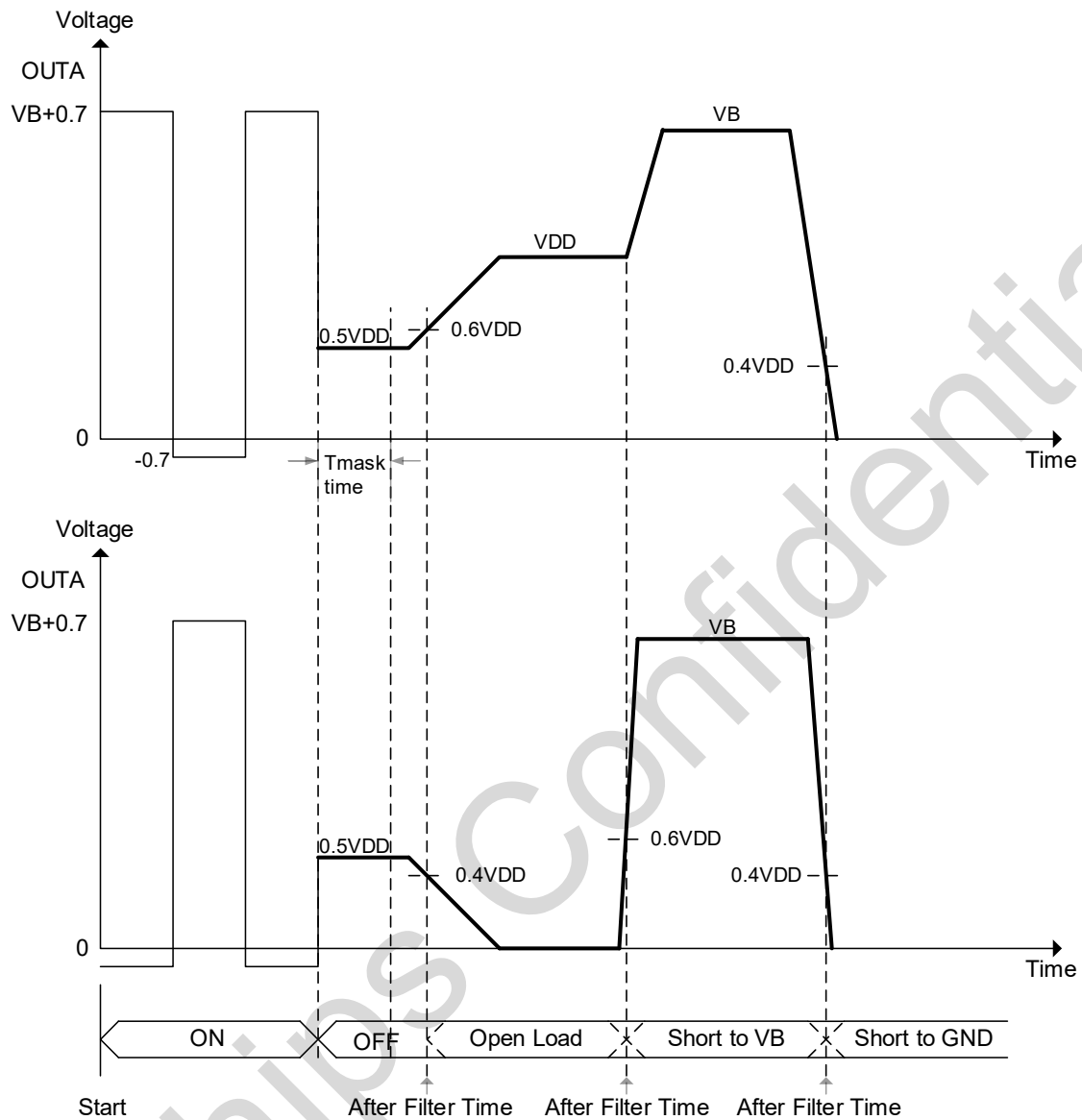
Figure 4-5 Low side driver diagnosis I-V relationship

4.4.6 Stepper motor driver OFF diagnosis (EN signal low and output in high impedance state)

In OFF condition Short to GND/Short to VB or Open Load condition is continuously detected through a deglitch filter of $125 \mu\text{s} \pm 25\%$, after Tmask masking time of $1\text{ms} \pm 25\%$ to filter ON/ OFF transition. To avoid false diagnostic due to motor residual movement, the stepper has to be disabled at least 40ms after the PWL signal has been disabled. A fault longer than deglitch time is latched. OFF state diagnostic fault can be overwritten by ON state fault. OFF state fault does not prevent the stepper from switching on. The latched fault is cleared on request.

4.4.7 Stepper motor driver ON diagnosis (EN signal high and output driven by input commands)

In ON condition when over current fault is detected and validated after digital filtering time Tdgc of $20 \mu\text{s} \pm 25\%$, the bridge is turned OFF and the fault is latched. The bridge is turned ON again by input command EN transition. The latched fault is cleared on request. Over current fault has higher priority with respect to OFF condition faults. Each Bridge has dedicated fault diagnosis detection coded by three bits.



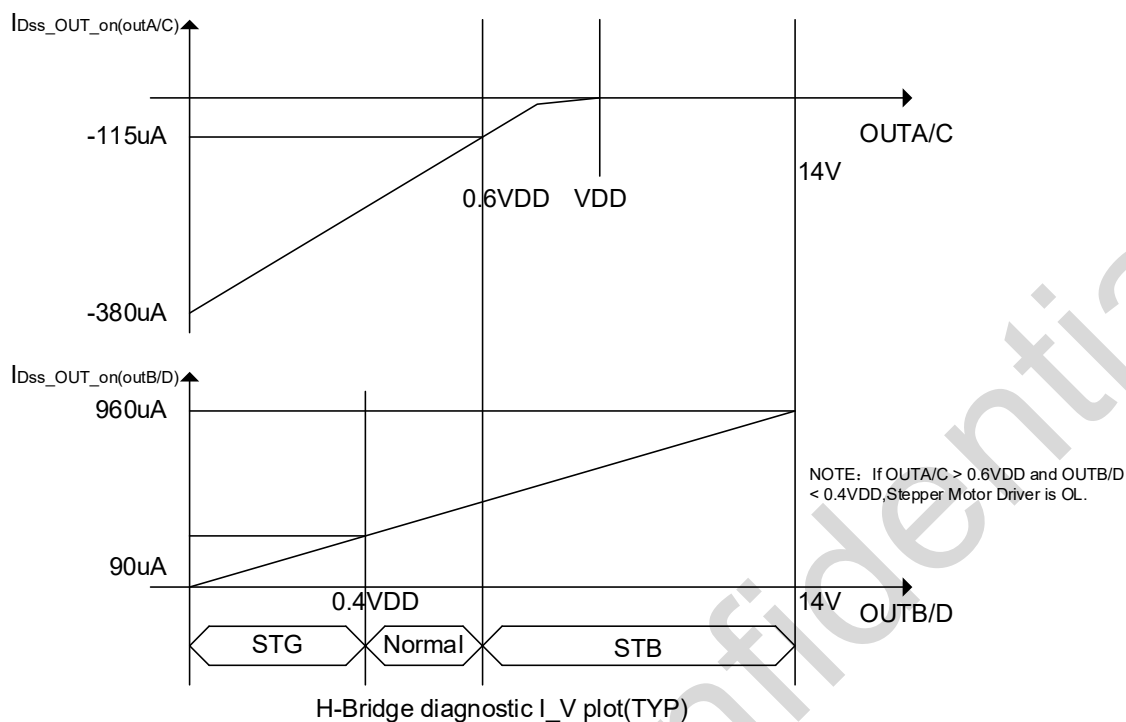


Figure 4-7 Stepper motor driver diagnosis I-V relationship

4.4.8 VRS diagnosis

VRS block enters diagnosis phase on request via SPI and then generates a Fault bit. If the fault exceeds the Tdgc filter time of $30 \mu s \pm 20 \%$, it is latched. The latched fault is cleared on request via SPI. The VRS diagnostic can only be activated when the phonic wheel is in stop condition.

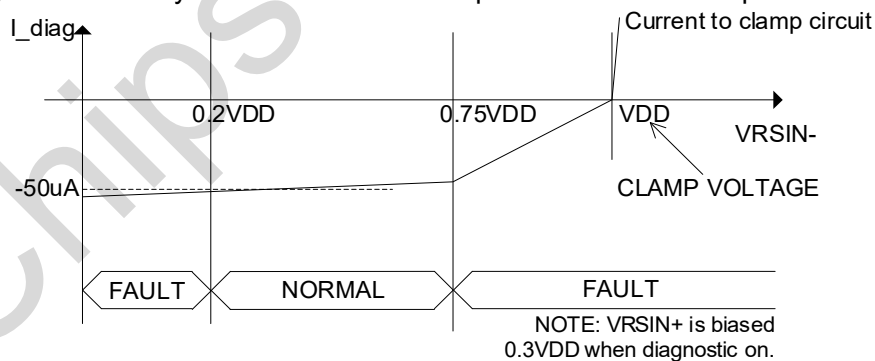


Figure 4-8 VRS diagnosis I-V relationship

4.5 VRS interface

4.5.1 Function characteristic

The flying wheel interface is an interface between the μP and the flying wheel sensor: it conditions signal coming from magnetic pick-up sensor or hall effect sensor and feeds the digital signal to microcontroller that extracts flying wheel rotational position, angular speed and acceleration.

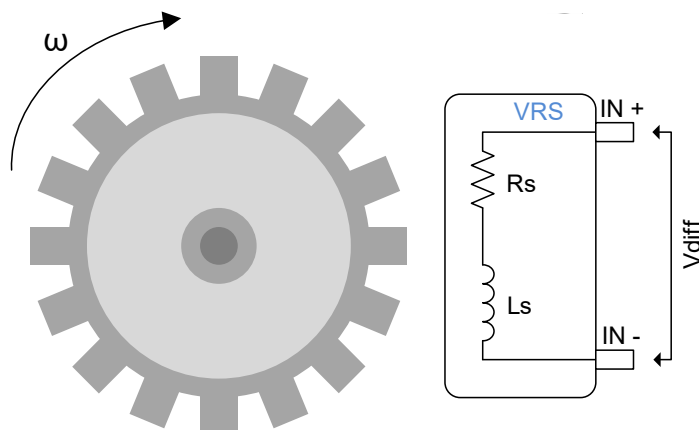


Figure 4-9 VRS typical characteristics

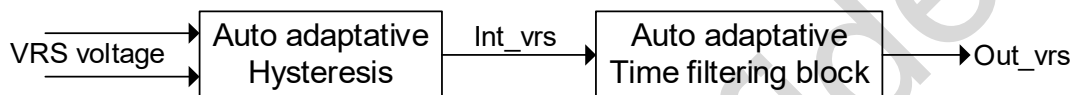


Figure 4-10 VRS interface structure

4.5.2 Auto-adaptive hysteresis

Input signals difference is obtained through a full differential amplifier; its output, DV signal, is fed to peak detection circuit and then to A/D converter implemented with 4 voltage comparator (5 levels PVi). Output of A/D is sent to Logic block (hysteresis selection Table 3-17) that implements correlation function between Peak voltage and hysteresis value; hysteresis value is used by square filtering circuit which conditions DV signal.

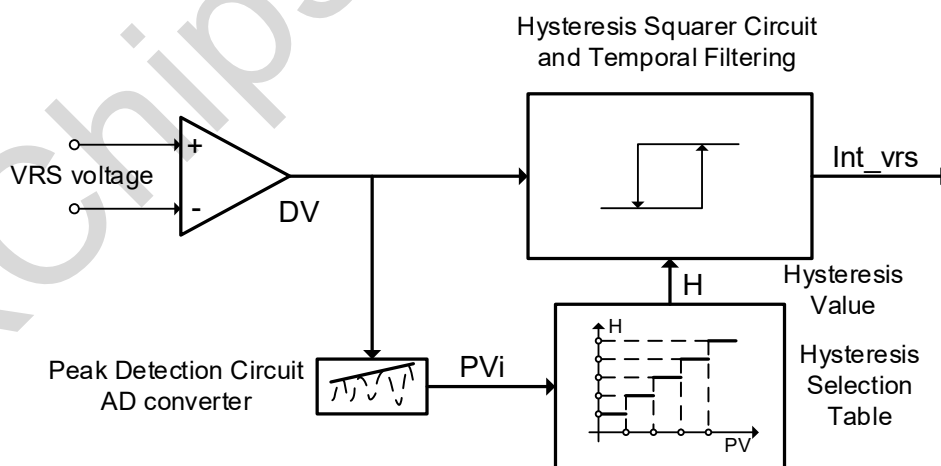


Figure 4-11 Auto-adaptive hysteresis block diagram

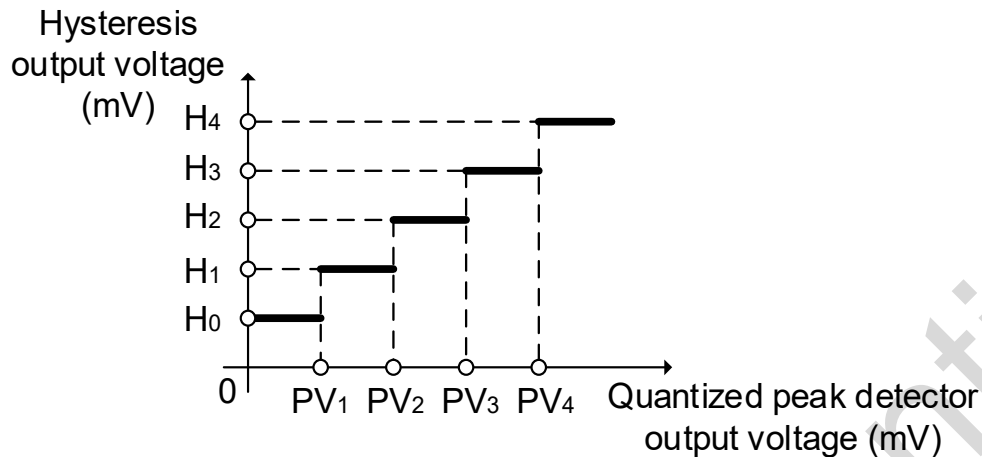


Figure 4-12 Hysteresis output voltage level

To the previous 5 levels $PV = [0 \ PV1 \ PV2 \ PV3 \ PV4]$ correspond to a set of 5 thresholds:

$H = [H0 \ H1 \ H2 \ H3 \ H4]$

The advised values for the previous defined vectors are:

$PV = [0 \ PV1 \ PV2 \ PV3 \ PV4] = [0, 930, 1600, 2300, 3000] \text{ mV}$

$H = [H0 \ H1 \ H2 \ H3 \ H4] = [100, 200, 340, 640, 1020] \text{ mV}$

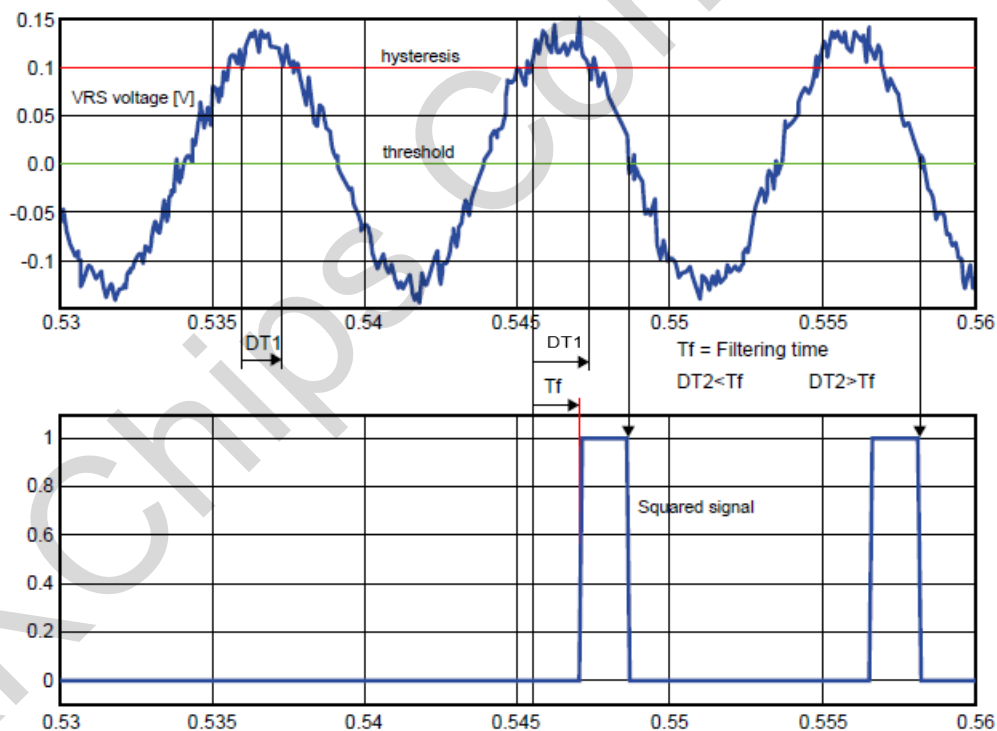


Figure 4-13 Input-output behavior of VRS interface

4.5.3 Auto-adaptative time filter

This characteristic is useful to set the best internal filter time as function of the input signal frequency.

The Tfilter time depends on the previous period duration T_n according to the following formula:

$$T_{\text{filter}}(n+1) = 1/32 * T_n \text{ if } T_n > T_{\text{filter}}(n)$$

The filtering time purpose is to filter very short spikes.

The digital filtering time is applied to internal squared signal (int_vrs), obtained by voltage comparators. The output of time filtering block is out_vrs signal.

The filtering time Tfilter is applied to int_vrs signal in two different ways:

- Rising edge: if int_vrs high level lasts less than Tfilter, out_vrs is not set to high level.
- In absence of any spikes during input signal rising edge out_vrs signal is expected with a delay of Tfilter time.
- Falling edge: the falling edge of int_vrs is not delayed through time filtering block: after falling edge for a time Tfilter any other transition on int_vrs signal is ignored

The initial value (Default) and maximum for Tfilter must be considered at RPM_min = 20 e.g. Tmax filter = 180 μs.

The minimum available value is Tmin filter = 2.8 μs.

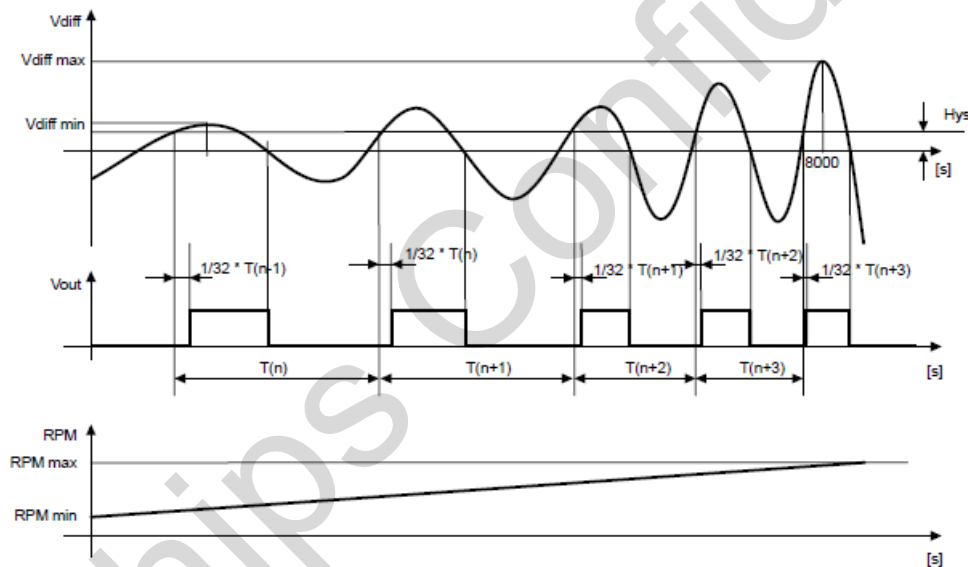


Figure 4-14 Auto-adaptative time filter behavior 1

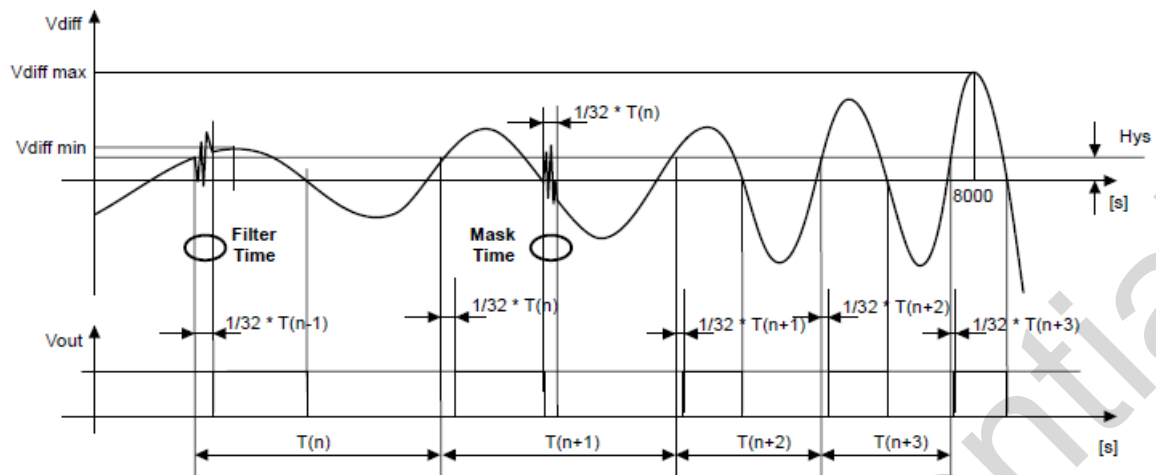


Figure 4-15 Auto-adaptative time filter behavior 2

4.6 Low side drivers

Low side drivers have a voltage slew rate control during switch-on/off phase to reduce emissions. The slew-rate control is achieved controlling the gate charging current and the behavior is described in [Figure 4-16](#) and [Figure 4-17](#).

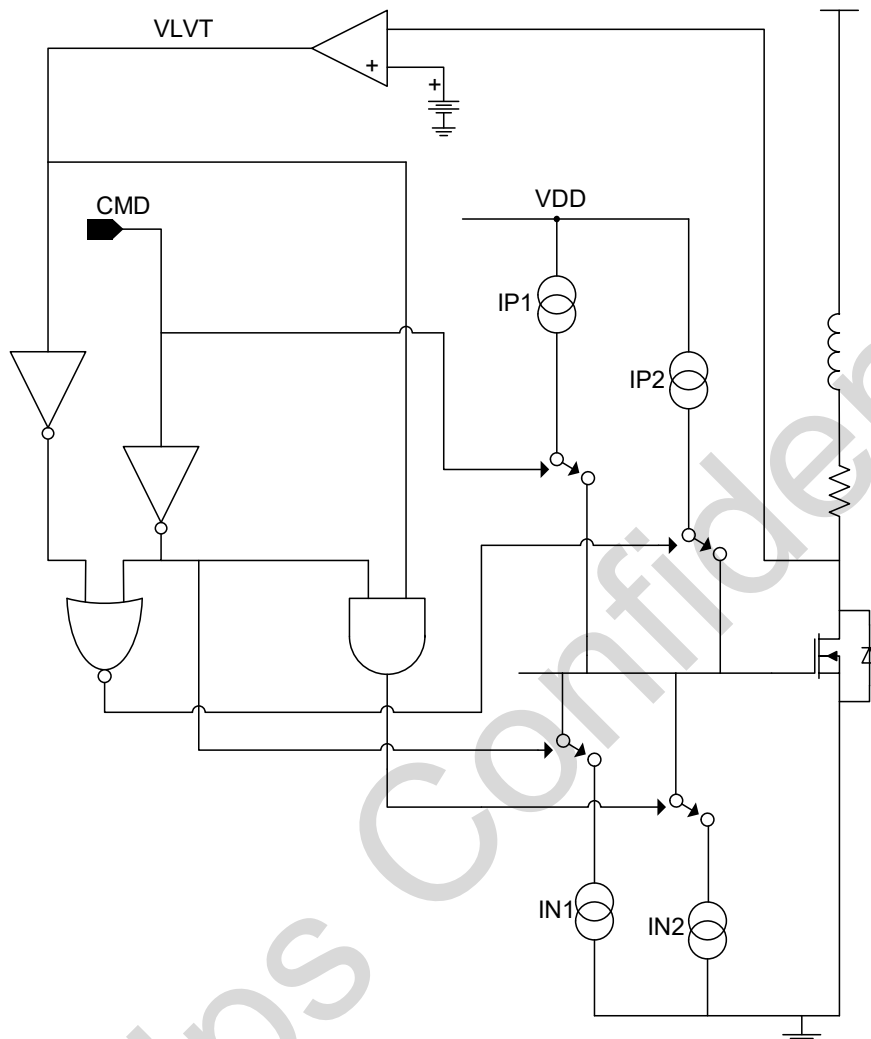


Figure 4-16 Low side drivers slew rate implementation

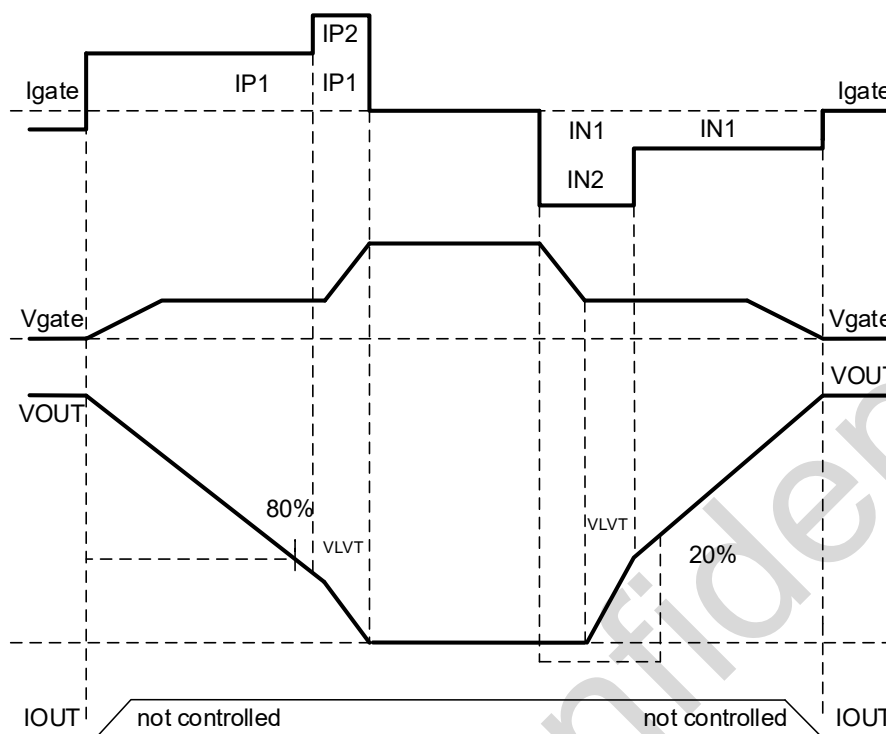


Figure 4-17 Low side drivers slew rate

At switch-on command the charging current is provided by current generator IP1 and is kept constant until the output voltage is decreased of roughly 80% of typical battery level. At this point the low side transistor is on and VLVT signal is set to logic 1 to connect IP2 current generator in parallel with IP1, completing the gate charge curve and providing maximum gate drive.

When the power transistor is switched-off the gate is discharged quickly using both IN1 and IN2 currents; as soon as the output voltage reaches roughly 20 % of the nominal battery voltage only IN1 is kept connected to complete the gate discharging.

In Table 4-8 the values for IPx and INx current generators are reported for each low side. As an example Figure 4-18 shows the resulting slew rate, in typical conditions, of O2H low side driver.

Table 4-8 Values for IPx and INx current generators for each low side

Low side	IP1	IP2	IN1	IN2	Unit
O2H	50	230	45	160	μA
INJ	105	140	100	160	μA
RLY	35	200	40	50	μA
TACH	55	140	40	42.5	μA
LED	37.5	140	27.5	12.5	μA

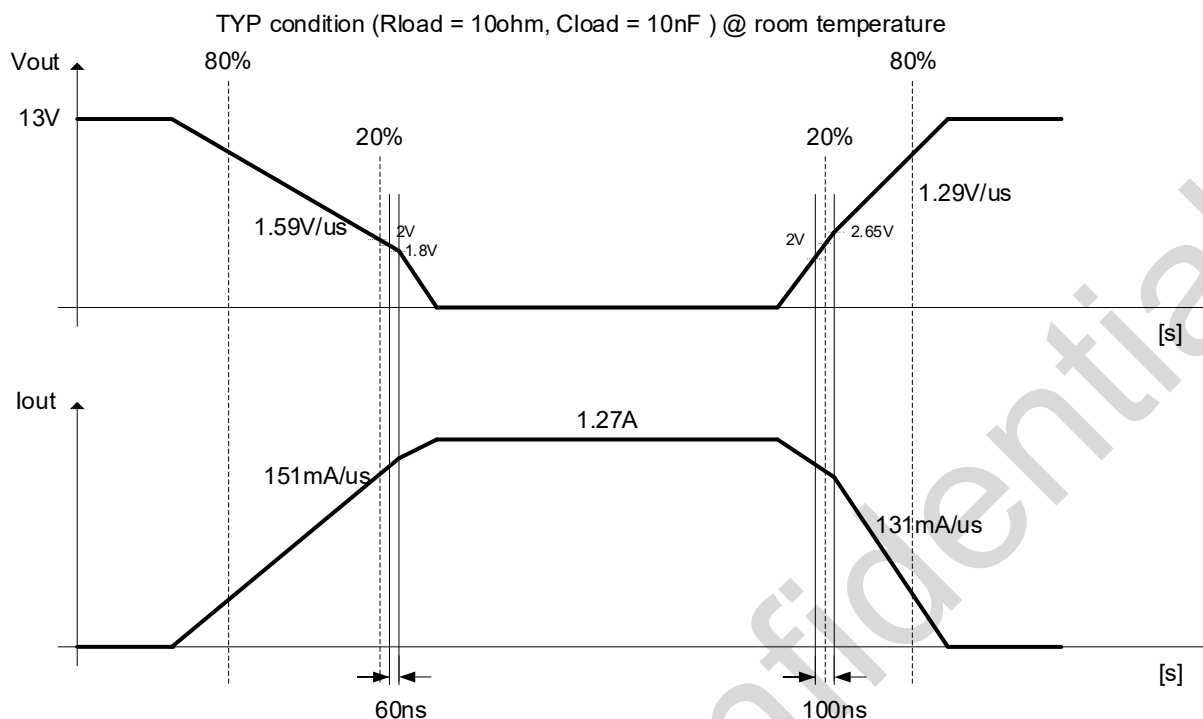


Figure 4-18 O2H low side driver slew rate

4.7 ISO serial line (K-LINE)

The ISO Serial Line is an interface containing one bidirectional line for communications between the μP and an external diagnosis tester or anti-theft device. In case of ground loss the outputs K_LINE gets in high impedance state and can withstand a negative voltage up to -18 V. Short circuit to Vb protection is provided: if the K_LINE pin is shorted to battery the output is switched off after a delay of T_filter_K_LINE. The fault bit is latched and is necessary to read and clear by SPI to turn on it again.

4.8 Stepper motor driver

Two H-Bridge for stepper motor driving (no half-bridge arrangement is possible). UN67C driving the motor movement is controlled through bit EN, DIR and PWM.

If the bit EN=1, the writing of bit PWM from 0 to 1 lead to the next step of the turn on sequence. The writing of bit PWM to 0 left unchanged the MOS of the bridge that are ON. The step is done only if the PWM bit goes from 0 to 1.

The order of the turn-on sequence is defined by the bit DIR.

In the below diagram the stepper motor operation.

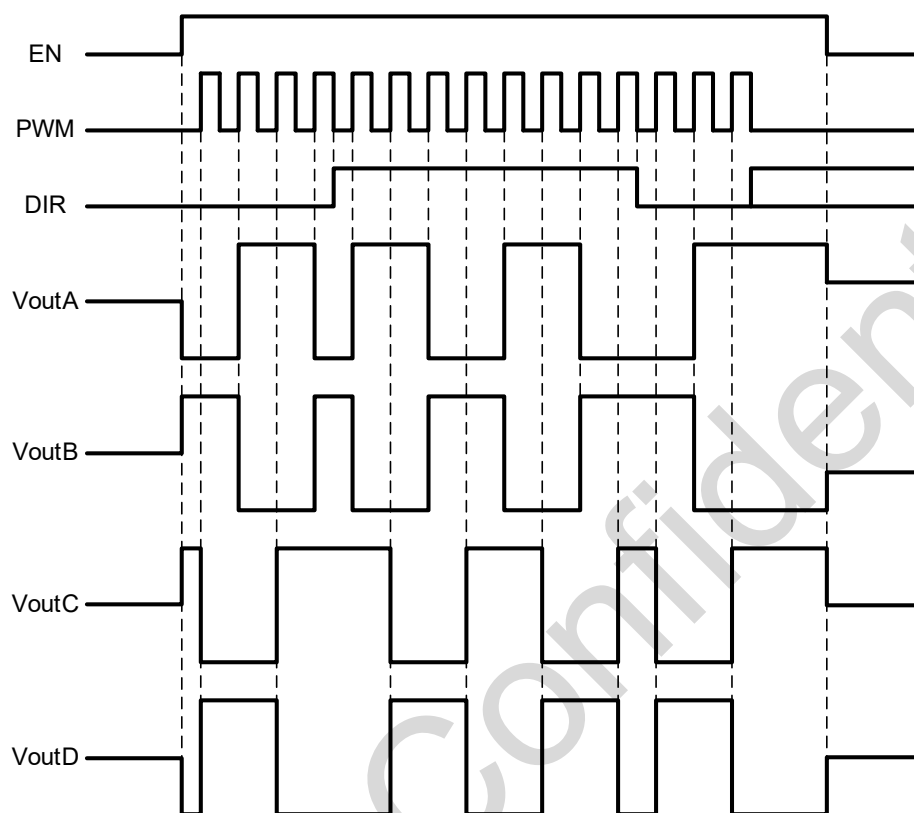


Figure 4-19 Stepper motor operations

5 Application circuit

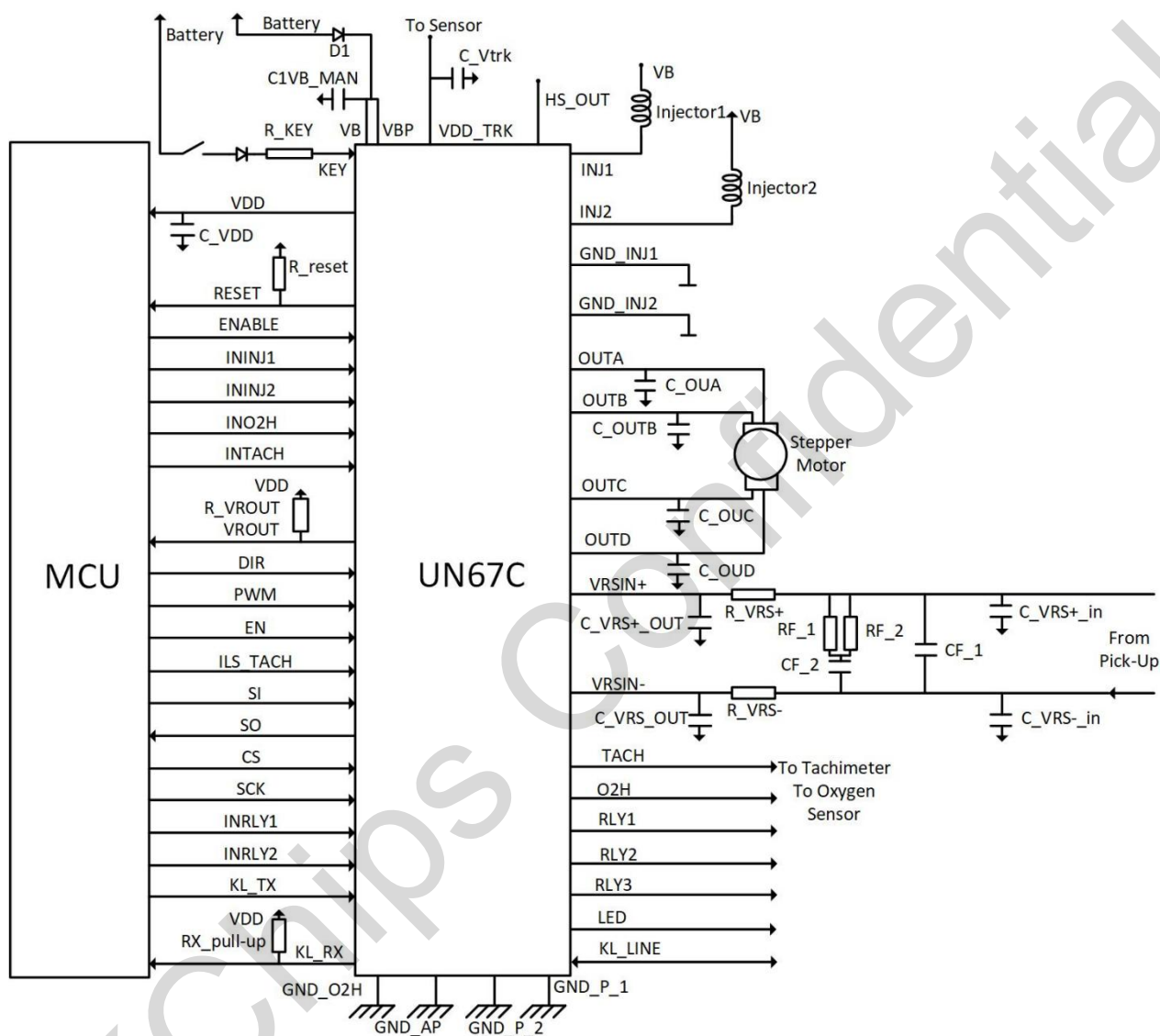


Figure 5-1 Application circuit

5.1 Bill of material

Table 5-1 Bill of material

Block	Component	Name	Usage	Min	Typ	Max	Unit
Supply	Capacitor	C1VB_MAIN	Bulk capacitor		220		μF
	Diode	D1	Reverse polarization protection diode				
Key	Resistor	R_Key	Current limiting resistor		20		kΩ
Reset	Resistor	R_reset	Pull-up resistor		1		kΩ

VDD	Capacitor	C_VDD value	Output capacitor (Ceramic or Tantalum)	1		10	μF
		C_VDD ESR				1.5	Ω
VDD_trk	Capacitor	C_Vtrk value	Output capacitor	0.1		1	μF
		C_Vtrk ESR		10		100	mΩ
VRS	Capacitor	C_VRS+_in	Filter Capacitor			100	pF
VRS	Capacitor	C_VRS-_in	Filter Capacitor			100	pF
VRS	Capacitor	C_VRS+_out	Filter Capacitor			470	pF
VRS	Capacitor	C_VRS-_out	Filter Capacitor			470	pF
VRS	Resistor	R_VRS+	Current limiting resistor			10	kΩ
VRS	Resistor	R_VRS-	Current limiting resistor			10	kΩ
VRS	Capacitor	CF_1	Filter Capacitor			470	pF
VRS	Capacitor	CF_2	Filter Capacitor			100	nF
VRS	Resistor	RF_1	Filter resistor			33	kΩ
VRS	Resistor	RF_2	Filter resistor			33	kΩ
VRS	Resistor	R_VROUT	Pull-up resistor	10			kΩ
K-Line	Resistor	RX_pull-up	Pull-up resistor			2	kΩ
Stepper	Capacitor	C_OUTA	EMI filter capacitor			10	nF
Stepper	Capacitor	C_OUTB	EMI filter capacitor			10	nF
Stepper	Capacitor	C_OUTC	EMI filter capacitor			10	nF
Stepper	Capacitor	C_OUTD	EMI filter capacitor			10	nF

Table 5-2 External Components Required

Pin	Symbol	Parameter	Value			Suggested part number
			Min	Typ	Max	
VTRK	C _{TRK}	External VTRK Capacitor	100 nF		1 μF	C1005X7R1C104K--0.1μF C1608X7R1H104K--0.1μF
VDD	C _{VDD5}	External VDD Capacitor	1μF		10μF	C2012X7R1E105K--1μF C1608X7R1C105K--1μF C3216X7R1H105K--1μF C3225X7R1E106K--10μF C3225X7R1C106K--10μF

Capacitor Legend:

1H->50V

1E->25V

1C->16V

X7R-> -40 to 125 °C +/-15%

K->-40 to 125 °C +/-10% 4

6 Package

ETQFP48 package information

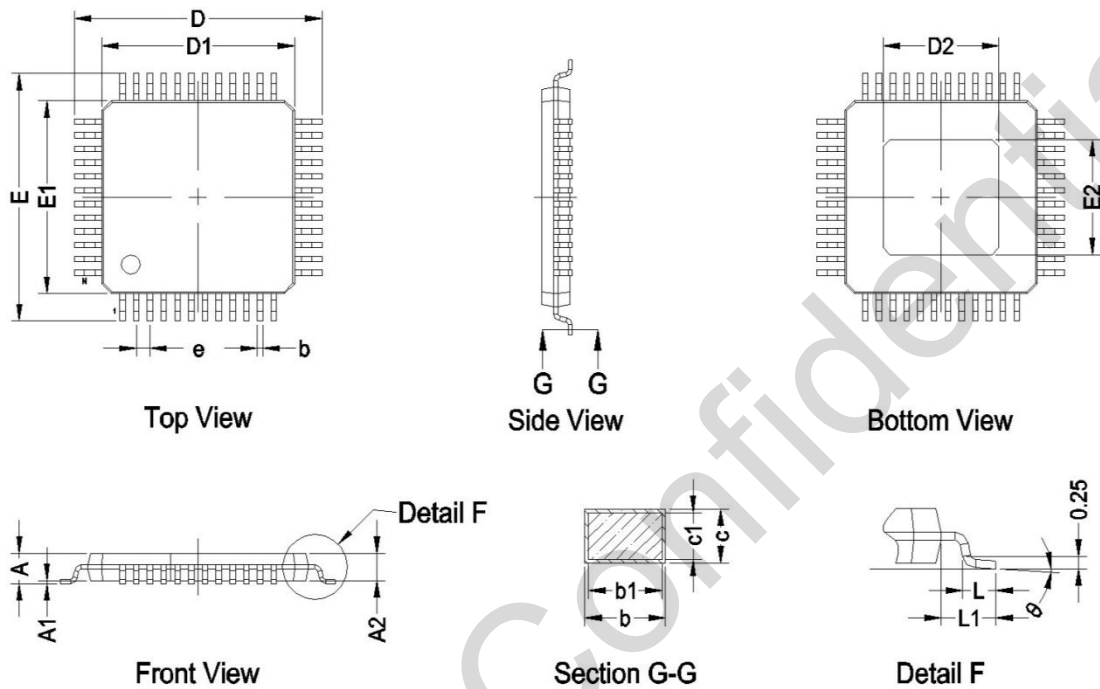


Table 6-1 ETQFP48 package mechanical data

Symbol	Min.	Typ.	Max.	Description
A	-	-	1.2	Total Thickness
A1	0.05	-	0.15	Stand Off
A2	0.95	1	1.05	Mold Thickness
b	0.17	0.22	0.27	Lead Width(Plating)
b1	0.15	0.20	0.25	Lead Width
c	0.10	-	0.2	L/F Thickness(Plating)
c1	0.09	0.127	0.16	L/F Thickness
D	8.80	9.00	9.20	Lead Tip Spacing
D1	6.90	7.00	7.10	Body Size(Length)
D2	3.15	-	4.20	EP Size(Length)
E	8.80	9.00	9.20	Lead Tip Spacing
E1	6.90	7.00	7.10	Body Size(Width)
E2	3.15	-	4.20	EP Size(Width)
e	-	0.50 BSC	-	Lead Pitch
L	0.45	0.6	0.75	Foot Length
L1	-	1.00 REF	-	Foot Print
N	-	48	-	Lead Quantity
θ	0°	-	8°	Foot Angle

7 Revision history

Date	Revision	Changes
2026/1/19	1.0	1. Initial version